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Price and Demand Responsiveness in the German Rental Housing Market

Interpretable Machine Learning on the RWI-GEO-RED Dataset (2007–2025)

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Project Paper Gestaltungsansätze der Advanced Data Analytics

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Submitted on: 7th September 2026

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1 Introduction

1.1 Motivation

The German rental housing market exhibits structural imbalances, with a shortage of housing in metropolitan areas contrasted by vacancies in peripheral regions. Germany lacked around 1.4 million dwellings at the end of 2024 (Günther, 2026), while asking rents in the 14 largest cities rose by around 44 % on average between 2016 and 2025, in Berlin and Leipzig by almost 70 % (Deutscher Bundestag, 2026). This divergence renders nationwide averages uninformative and calls for models that explain rents from individual property and location characteristics.

Hedonic price theory (S. Rosen, 1974) decomposes the rent into implicit attribute prices via linear regression, whose coefficients allow a direct economic interpretation. Its linearity and additivity assumptions, however, fail to capture the nonlinearities and spatial interaction effects of real estate markets. Tree-based machine learning methods capture these structures automatically and achieve higher predictive accuracy, but obscure the internal relationships. Mullainathan and Spiess (2017) frame this tension as a conflict between *prediction* and *inference*, a gap between "interpretable but imprecise" and "accurate but opaque".

The present work addresses this trade-off with two interpretable "glassbox" architectures based on generalized additive models (GAMs), the Explainable Boosting Machine (Nori et al., 2019) and IGANN (Kraus et al., 2024). Their additive structure expresses the contribution of each feature as a readable *shape function*, without the linearity assumption of the hedonic model. Lorenz et al. (2023) show that interpretable methods achieve a predictive performance on real estate markets similar to that of black-box models. To assess the robustness of the feature effects, the glassbox models are contrasted with post-hoc TreeSHAP explanations of the black-box models (Lundberg and Lee, 2017; Lundberg, Erion, et al., 2020).

Beyond the price, this work models the marketing duration (*time-on-market*) as a second target variable, since asking rents alone paint an incomplete picture of market valuation. A high asking price need not find market clearance. The empirical basis is a

raw dataset of more than 27 million German rental listings. This leads to the following research question.

> *Are the rental price and the time-on-market of rental apartments in Germany associated with the same characteristics, and to what extent do interpretable models (EBM, IGANN) and post-hoc explanations (TreeSHAP on XGBoost/LightGBM) provide consistent insights into the feature effects?*

1.2 Goals and Structure of the Paper

Building on the research question derived in Section 1.1, this work pursues three complementary goals.

1. **Predictive model comparison (dual-target).** Black-box models (XGBoost, LightGBM), the tabular foundation model TabICL, and intrinsically interpretable glassbox models (EBM, IGANN) are systematically compared on two target variables, rent per square meter (Target A) and time-on-market (Target B). A regularized linear regression (ridge) serves as a hedonic baseline to quantify the predictive added value of the nonlinear methods.

2. **Spatial feature engineering.** The property-level attributes are extended by a spatial dimension. Infrastructure densities and distances to the district center, computed from OpenStreetMap raw data, operationalize the concept of the "15-minute city". The technical implementation is documented in Section 4.2.

3. **Explainability and consistency check.** Beyond predictive accuracy, the models trained in Goal 1 are compared at the level of their feature effects (shape functions) along two axes, (a) between the two target variables (identical feature set, different targets) and (b) between the model types (intrinsic glassbox explanations versus post-hoc attributions via TreeSHAP).

Chapter 2 outlines the domain context, the theoretical framework, and the methodological scope of the work. Chapter 3 positions the study within the state of research on hedonic price models, ML-based valuation, interpretable machine learning, and time-on-market. Chapter 4 describes the dataset, the preprocessing and feature engineering, the model specifications, and the evaluation methodology. Chapter 5 presents the model performance and the shape function comparison for both target variables, and Chapter 6 interprets the findings, discusses limitations, and provides an outlook.

2 Foundations and Problem Description

2.1 Domain Context

The German rental market comprises two price segments that follow distinct dynamics: existing tenancies and new rentals. Rents in existing contracts are subject to a statutory increase cap of 20 % within three years (§ 558 BGB) (*§ 558 Mieterhöhung bis zur ortsüblichen Vergleichsmiete* n.d.), which dampens their adjustment to market conditions. New rentals are largely unconstrained and reflect current supply-and-demand conditions directly. As a result, the two segments have diverged by 43 %, as Bundesinstitut für Bau-, Stadt- und Raumforschung (2026) reports. Since the ImmoScout24 listings underlying this study capture exclusively the new rental segment, the observed asking rents reflect the unattenuated market dynamic, but also limit the viewpoint to only a part of the market reality.

2.2 Theoretical Framework: Price Formation and Demand Responsiveness

The research question introduced in Chapter 1 asks whether rental price and time-on-market are governed by the same underlying feature structure. Answering this question presupposes that both quantities can be meaningfully expressed as a function of the same set of housing attributes. This paper rests on two conceptual premises that establish this link and motivate the further analytical approach.

First, a housing unit is treated as a bundle of valued attributes, so that its rental price can, in principle, be decomposed into the contribution of each individual attribute rather than treated as a single latent quantity. Second, the time a listing remains on the market is treated as the observable outcome of a matching process between the attributes offered and prevailing demand, rather than as a price signal in its own right. Taken together, both premises imply that each attribute exerts an identifiable influence on both target variables – an influence that can be isolated and expressed as a separate, estimable function. These

functions, referred to as shape functions throughout this paper, form the basis of the comparison conducted in Section 5.4.

2.3 Methodological Scope

The premises established above imply that "same feature structure" can be assessed at the level of individual attributes rather than as a global property of a model. Both targets, price per square meter (Model A) and time-on-market (Model B), are therefore predicted from an identical feature matrix, ensuring that any observed differences between the two targets originate from how the attributes act on each outcome, not from differences in the information available to the models.

Aggregate feature-importance rankings or coefficient magnitudes would be insufficient here, as they collapse a feature's direction, functional form, and value range into a single figure. The shape function preserves this structure and is therefore adopted as the unit of comparison.

The comparison is carried out along two axes. First, a cross-method check examines whether different models trained on the same target arrive at consistent feature-level effects, establishing whether an observed pattern reflects the data or is an artifact of a particular model class. Second, a cross-target comparison contrasts the shape functions obtained for price and time-on-market with respect to sign, trend, and functional form, which constitutes the main analytical result of this paper.

The scope of this comparison is explicitly predictive rather than causal. The estimated feature effects are predictive associations, not causal parameters.

3 Related Work

3.1 Hedonic Pricing Models in Real Estate

The theoretical foundation of real estate valuation is hedonic price theory, formalized by S. Rosen (1974). It conceives of housing as a bundle of utility-bearing attributes such as living space, building age, location, and amenities. The observed market price equals the sum of the implicit prices (shadow prices) of these attributes. In the first stage of the model, the hedonic price function is estimated and its partial derivatives yield the implicit prices. The second stage aims to recover the structural demand and supply functions from these prices.

The approach transfers directly to the rental market. The rent is not overlaid by financing conditions or speculative appreciation components and approximately reflects the willingness to pay for the housing service itself, which motivates rent per square meter (Target A) as the primary target variable of this work.

Traditionally, the hedonic model is estimated by linear regression (OLS) in a semi-logarithmic specification. Its coefficients can be read directly as marginal willingness to pay under *ceteris paribus* conditions, complemented by an established inferential toolkit of p-values and confidence intervals (Sirmans, Macpherson, and Zietz, 2005). These virtues come with methodological limits. OLS imposes an a priori functional form, whereas relationships in housing markets are frequently nonlinear, for example the saturation effect of additional living space. Moreover, the value of an attribute is spatially unstable, and adding many interaction terms by hand quickly leads OLS into the curse of dimensionality. Spatial econometrics addressed this spatial instability with spatial lag, spatial error, and spatial Durbin models (Anselin, 1988; LeSage and Pace, 2009), but extends the linear paradigm only by a spatial term and leaves nonlinearity and high-dimensional interactions unsolved.

Beyond the functional form, the second stage of the hedonic model, the recovery of the demand function, is not identified without restrictive assumptions or external instruments (Brown and H. S. Rosen, 1982; Ekeland, Heckman, and Nesheim, 2004). For the present work, this implies a restriction to the first stage. Feature effects are interpreted

as predictive associations rather than structural-causal demand parameters. The predictive accuracy sacrificed by the linear paradigm motivates the shift to machine learning methods, which Section 3.2 addresses.

3.2 Machine Learning for Real Estate Price Prediction

To overcome the weaknesses of linear models, the real estate literature has moved toward machine learning. Tree-based ensemble methods such as Random Forest (Breiman, 2001), XGBoost (Chen and Guestrin, 2016), and LightGBM (Ke et al., 2017) have become the de facto standard. Decision trees partition the feature space in a data-driven way and thereby capture nonlinearities and spatial interaction effects without a prespecified functional form.

Comparative studies confirm their predictive advantage over OLS across markets and datasets (Park and Bae, 2015; Ho, Tang, and Wong, 2021; Mullainathan and Spiess, 2017). The majority of these studies focus on purchase or transaction prices, while the rental market is studied far less frequently. This work addresses that gap.

On tabular, heterogeneous data, the typical format of real estate listings, tree ensembles outperform neural networks (Grinsztajn, Oyallon, and Varoquaux, 2022), which justifies the focus of the present work on XGBoost and LightGBM. This work additionally tests TabICLv2 (Qu et al., 2026), a tabular foundation model that transfers the in-context-learning approach of TabPFN (Hollmann et al., 2025) to substantially larger training sets.

Despite their predictive superiority, classical ML models hit the conflict between *prediction* and *inference* introduced in Section 1.1 (Mullainathan and Spiess, 2017). A model such as XGBoost consists of hundreds of serially linked trees, so the isolated marginal effect of an individual feature, the core of hedonic theory, remains invisible. Two methodological ways out exist. Post-hoc explanations approximate an already trained black-box model after the fact, while intrinsically interpretable glassbox models guarantee transparency by design. Both paths are developed in Section 3.3.

3.3 Interpretable Machine Learning and Generalized Additive Models

Rudin (2019) argues that in high-stakes decision domains intrinsically interpretable models are preferable to post-hoc explained black boxes, because post-hoc explanations remain

approximations of the original model. Lorenz et al. (2023) are the first to show systematically that intrinsically interpretable models reach a predictive performance close to that of black-box models on real estate markets while keeping the feature effects fully transparent. Their study forms the central point of departure for the present work. Kruschel et al. (2025) complement it with an information-systems perspective.

Intrinsically interpretable models are rooted in generalized additive models (Hastie and Tibshirani, 1986). A GAM replaces the linear terms of the regression with flexible, nonlinear shape functions and models the transformed expectation of the target variable additively as $g(E[y]) = \beta_0 + f_1(x_1) + \cdots + f_k(x_k)$. The contribution of each feature thus remains readable as a separate, visualizable curve, while the linearity assumption of OLS is lifted.

The Explainable Boosting Machine (Nori et al., 2019) estimates each shape function via cyclic gradient boosting of shallow trees restricted to a single feature and adds automatically selected pairwise interaction terms (GA^2M) (Lou et al., 2013). IGANN (Kraus et al., 2024) estimates the shape functions through extreme learning machines in a boosting procedure and has been less explored in the real estate literature.

The complementary branch consists of post-hoc methods. The standard is SHAP (Lundberg and Lee, 2017), which assigns each feature an additive contribution to an individual prediction based on Shapley values, with TreeSHAP (Lundberg, Erion, et al., 2020) providing an exact and efficient computation for tree ensembles. SHAP values are, however, an approximation of the model's logic, and with correlated features the attributions can become ambiguous. Precisely here lies the methodological contribution of the present work. The consistency check between the intrinsic shape functions (EBM, IGANN) and the post-hoc explanations (TreeSHAP on XGBoost/LightGBM) tests whether both approaches lead to concurring feature effects. Divergences point to interaction effects or modeling artifacts (Section 4.6).

3.4 Time-on-Market Research

Unlike financial assets, housing is illiquid, and letting an apartment takes days to months. A high asking price need not clear the market, so asking rents in isolation remain an incomplete indicator of market valuation. The marketing duration (time-on-market, ToM) quantifies this demand response and becomes a target variable in its own right.

Theoretically, the ToM is anchored in search theory (Haurin, 1988; Anglin, Rutherford, and Springer, 2003; Haurin et al., 2010). A higher asking price increases the revenue per transaction, but lowers the arrival rate of suitable prospects and prolongs the expected

search duration. A short ToM signals market-conform pricing, while a long ToM points to overpricing, low property attractiveness, or market saturation. Empirically, asking prices above market value prolong the marketing duration disproportionately, with a convex rather than linear relationship (Anglin, Rutherford, and Springer, 2003), and atypical properties exhibit longer times on market.

Most ToM research, however, stems from the purchase segment, while the rental market is far less studied although the search-theoretic logic transfers directly. The present work addresses this gap and models the ToM as a separate target variable (Target B). Modeling both targets on an identical feature set enables the central question of whether the same characteristics shape price and marketing duration or whether systematic asymmetries exist. High-end amenities, for instance, may raise the price but prolong the time on market. This question is evaluated through the direct comparison of the shape functions (Section 5.4).

4 Research Method

4.1 Dataset

This work draws on the dataset *RWI-GEO-RED v14* (Real Estate Data for Germany), provided as a Scientific Use File (SUF) by the Research Data Centre Ruhr (FDZ Ruhr) at the RWI Leibniz Institute for Economic Research (RWI – Leibniz-Institut für Wirtschaftsforschung and ImmobilienScout24, 2026; Heuer and Thiel, 2026). The raw data originate from ImmobilienScout24 (IS24), the largest German real estate platform. The FDZ divides the data into four subsets (houses and apartments, each for sale and for rent). This work uses the subset *apartments for rent* (DOI `10.7807/immo:red:wm:suf:v14`, raw file `WM_SUF_mitText.csv`, about 9.5 GB including free-text fields). According to the FDZ, the data constitute the *full population* of all residential listings on the platform rather than a drawn sample. The substantive coverage is therefore determined primarily by the platform coverage itself.

The dataset records *asking rents* at the listing level (so-called spells), not the finally contracted transaction rents, since the advertised price is non-binding. The dependent quantity is the net cold rent (`mietekalt`), the price for the pure living space excluding utility, heating, and other costs, from which the area-normalized target variable `mietepro_m2` (Target A) is derived. Since advertised rents in tight markets can be capped by regulation (rent brake, § 556d BGB), high willingness to pay is potentially underrepresented, which has to be considered when interpreting the price effects.

Temporally, the data form a longitudinal panel from January 2007 to December 2025 with monthly resolution. The variables `ajahr`/`amonat` mark the start of a listing, `ejahr`/`emonat` its end. The derived listing duration (`laufzeittage`) forms the basis of Target B (time-on-market). The listing duration is a *proxy* for the actual marketing duration, because the deactivation of a listing usually, but not necessarily, indicates a successful letting. Across the final dataset, the two target variables are only weakly correlated ($r = -0.119$ between `mietepro_m2` and the log-transformed listing duration), a first indication that asking price and demand response capture distinct dimensions of market valuation. After the initial filter on valid cold rents (`mietekalt` > 0), the raw stock

comprises 27,345,526 rental listings. After full cleaning (deduplication to one observation per property, plausibility filters, and treatment of the FDZ censoring codes), an effective working set of 12,034,680 observations remains. Section 4.2 documents the corresponding pipeline.

Spatial referencing relies on the 1×1 km INSPIRE grid (`ergg_1km`, format `East_North`, ETRS89-LAEA/EPSG:3035). Exact address coordinates are not included in the SUF. In addition, the FDZ provides administrative BKG keys (2019 territorial status) for the district (`kid2019`) and the municipality (`gid2019`) as well as the functional labor market region (`erg_amd`) based on commuting linkages (Kosfeld and Werner, 2012). These variables form the basis of the spatial feature engineering (Section 4.2). For data protection reasons, grid cells with very few households (for apartments, fewer than 5 households or fewer than 10 households with a listing share of at least 50 %) are anonymized at the grid level. District and municipality keys are retained. This selectively limits the spatial resolution in sparsely populated, predominantly rural regions and explains part of the sample loss documented in Section 4.2.

For model training, a 10 % sample stratified by listing year (`ajahr`) is drawn from the final working set and used uniformly for all models. The stratification preserves the temporal distribution of the full dataset without bias. The only exception is the in-context-learning model TabICL, whose usable training set is additionally limited by the available GPU memory. Sampling and data splitting are documented in Section 4.2.

With a self-reported market share of around 50 % of all properties offered for sale or rent, IS24 is the largest German platform (Georgi and Barkow, 2010). Within this segment the data constitute a full census, yet the dataset covers only the *publicly listed* supply segment. Lettings without a public listing, for example by large portfolio holders, municipal and cooperative allocation, or informal networks, are underrepresented. This coverage limitation is taken up in Chapter 6.

4.2 Data Preparation and Feature Engineering

The preparation of the raw data follows a three-stage, fully reproducible notebook pipeline (01 Loading → 02 Feature Engineering → 03 Spatial Engineering). Guiding principles are the consistent avoidance of target leakage, the preservation of missing values as information until model training, and the transparent documentation of all filtering decisions.

4.2.1 Data Import and Big Data Processing

The raw file of around 9.5 GB is read with the lazy API of the Polars library (`scan_csv`). Operations are executed only at materialization (`collect()`), which avoids memory peaks. Already during the scan, the data are filtered to valid cold rents (`mietekalt` > 0). Persistence uses the column-oriented Parquet format, which allows selective column reads and drastically reduces file size. This step yields 27,345,526 rows.

4.2.2 Treatment of Missing Values

The FDZ codes missing values not as `null` but as negative integers (-5 "implausible", -6 "old variable", -7 "not specified", ..., -11 "anonymized"). All numeric columns with values below zero are therefore systematically converted into true `null` values to rule out logical errors in model training (such as "minus five rooms"). The preparation pipeline itself does not impute. Numeric and categorical missings remain `null`, so that the absence of a value is preserved as information. Their treatment takes place only in model training. After the train-test split, all models uniformly receive a median imputation for numeric and a mode imputation for categorical features. The imputers are fitted exclusively on the training data to rule out leakage from the test set. Binary amenity features are interpreted as 0 in line with the FDZ recommendation, because positive features are actively advertised in listings and their absence approximately corresponds to "not present".

4.2.3 Deduplication at the Property Level

A physical property can appear as several listings (spells), for example through pausing and reactivation or minor adjustments by agents. Frequently relisted and thus systematically harder-to-let properties would distort the sample and both target variables. The dataset is therefore reduced to exactly one observation per property (`obid`). As a deterministic, reproducible selection criterion, the observation with the highest value of the FDZ duplicate identifier `dupID_gen` is kept per property (`sort` $\rightarrow$ `unique(keep="last")`). The criterion serves solely the unambiguous selection and is not interpreted substantively. A limitation remains in that `dupID_gen` classifies the *relationship* between duplicate spells (among other things temporal distance and attribute deviation), not their chronological order. The kept listing is therefore not necessarily the most recent one, so that a residual heterogeneity remains in the spell selection.

4.2.4 Plausibility Filters and Construction of the Target Variables

Subsequently, realistic rental market value ranges are enforced. Admissible are cold rents between 50 and 10,000 euros, living areas between 10 and 500 m^2, and construction years from 1850 up to `ajahr` + 2 (new-build tolerance). Target A is computed as rent per square meter `mietepro_m2` = `mietekalt`/`wohnflaeche` and capped by global winsorization at the 1 %/99 % quantiles (around 26.50 euros/m^2). The capping is deliberately applied globally across all years rather than per year, because a year-wise clipping would have destroyed the absolute scaling over the 19-year observation period and smoothed away the macroeconomic rent growth. Target B (`laufzeittage`) is transformed into `log_laufzeittage` via $\log(1 + x)$ to address its pronounced right skew.

4.2.5 Feature Construction and Leakage Prevention

Because of the perfect multicollinearity `alter_bauwerk` = `ajahr` − `baujahr` (age-period-cohort problem), `baujahr` is discarded for training. The feature `ajahr` carries the macroeconomic time trend, while the building age `alter_bauwerk`, clamped to values ≥ 0, carries the physical wear. Listing year `ajahr` and listing month `amonat` enter the training as numeric features. This encoding allows the linear baseline to learn the yearly trend directly as a slope, while tree-based models and GAMs partition the time axis in a data-driven manner anyway. A cyclical encoding of the month (sine/cosine) is not used, since tree-based models can only approximate cyclical encodings through step-like splits. A limitation remains in the year-end jump of the numeric month scale (December = 12, January = 1), which matters only for the linear baseline.

In addition, several features are derived. These include binary information flags from columns with a high missing share (`hat_parkplatz`, `wurde_modernisiert`, `hat_energieausweis`), a regex parser for the availability period `freiab_monate` as an urgency indicator (capped at 0 to 24 months), the `completeness_score` as the completion rate of a listing, and the logical cap `schlafzimmer` ≤ `zimmeranzahl`. The `completeness_score` is computed strictly *before* any imputation and exclusively over the 34 columns retained after the initial column selection (not all 91 raw columns). It thus measures the coverage of the analysis-relevant column space, not the general diligence of the advertiser.

To avoid target leakage, all post-publication variables are removed from the price model (Target A), namely web analytics quantities (`hits`, `click_*`), the listing end (`ejahr`/`emonat`), and the duration itself, since this information is unknown at listing time. The deposit is deliberately not parsed, because it is typically stated as a multiple of the

cold rent and thus has a direct leakage link to the target. Finally, strictly nominal codes (`heizungsart`, `objektzustand`, `kategorie_Wohnung`, `blid`, `erg_amd`) are defensively cast to strings so that their encoding in model training is explicit and model-appropriate. The linear baseline, IGANN, and the tabular foundation models receive one-hot encoded features, which rules out learning arbitrary ordinalities (such as "Bavaria = 9 > Berlin = 3"). The high-cardinality feature `erg_amd` (141 levels) is removed for this model group to avoid a dimensionality explosion. The tree-based models receive ordinally encoded categories, which is unproblematic since trees can isolate individual categories through repeated threshold splits. EBM processes the string categories natively and requires no encoder. All encoders are fitted exclusively on the training data. Features with untenable missing rates or near-zero variance are removed (`energieeffizienzklasse` with 88.7 % missing, `ausstattung` with 52.5 %, `rollstuhlgerecht` and `denkmalobjekt` with 0.01 % positives, `immobilientyp` as a constant).

4.2.6 Spatial Feature Engineering

From the grid identifier `ergg_1km`, the metric coordinates (EPSG:3035) are extracted and shifted by +500 m per axis to correct from the lower left cell corner to the cell center. From a 4.7 GB OSM dump for Germany, more than 430,000 points of interest (POIs) are extracted via Pyosmium and persisted as GeoParquet. The distance and density computation avoids a quadratically scaling cross join through a k-d tree from the SciPy library (`scipy.spatial.cKDTree`, $O(n \log n)$) (Virtanen, Gommers, Oliphant, et al., 2020). Per property, the distance to the nearest POI and the POI density within a 1 km radius are computed, each for public transport and supermarkets as an operationalization of the "15-minute city". In addition, `dist_to_center` is constructed as the Euclidean distance to the geometric centroid of the district (BKG VG250). The geometric centroid is preferred over error-prone OSM town hall tags, and the district geometries are deduplicated on `kid2019` before the join to avoid a row explosion. The originally planned park features (`dist_park`, `dens_park`) were discarded because the OSM query (`leisure=park`) systematically missed large forest and recreational areas (median density 0, maximum distance 66 km). The final dataset (`AUTO_final_enriched.parquet`) comprises 11,505,472 observations. The reduction of around 529,000 relative to the cleaned set results from excluding properties with anonymized or invalid grid cells without a coordinate (cf. Section 4.1).

4.2.7 Data Splitting and Hyperparameter Optimization

Model training is carried out uniformly for all models on a 10 % sample of the final dataset stratified by listing year (around 1.15 million observations). The stratification preserves the yearly distribution and ensures that the subsequent temporal cut is made on a representative time series. The only exception is the in-context-learning model TabICL, whose usable context size is limited by the available GPU memory. It operates on a random subset of 100,000 observations of the training set. A random train-test split would create *temporal leakage* with longitudinal data, because the model would learn from future prices to evaluate past ones. Evaluation therefore uses a hard temporal cut (out-of-time split). The years 2007 to 2022 form the training set (968,884 observations or 84.2 %), the years 2023 to 2025 the test set (181,662 observations or 15.8 %). Cutting after 2022 also ensures that the structural break of 2022 (the interest rate reversal at the end of the COVID-19 pandemic) is contained in the training data and the models do not have to extrapolate the changed price level as an entirely unknown future. Hyperparameter tuning is performed with Optuna (Akiba et al., 2019) on a separate 1 % sample, likewise stratified by listing year and restricted to the training period, in a 3-fold cross-validation. Since the data are ordered by listing year and the folds are formed without shuffling (`KFold` with `shuffle=False`), they correspond to chronological segments. A remaining limitation is the absence of spatially blocked cross-validation. Spatially neighboring observations are correlated across the training and test sets, so the measured predictive power for entirely unknown regions is likely to be slightly overestimated. This limitation is discussed in Chapter 6.

4.3 Models

This study employs a deliberate portfolio of six regression models organized across three tiers of increasing predictive flexibility and decreasing structural transparency. The same feature matrix $\mathbf{x} \in \mathbb{R}^{32}$, comprising 16 numerical, 5 categorical, and 11 indicator features (ten binary flags and the three-level indicator `haustier_erlaubt`), is applied identically to both target variables. This design serves two purposes: it enables a fair, controlled comparison of predictive performance across model families and, more importantly for the scientific contribution of this work, it allows a direct side-by-side comparison of learned feature effects between Model A and Model B through the shape functions produced by the inherently interpretable models in Section 5.4. The breadth of the model portfolio furthermore reflects a deliberate learning objective: by covering the full spectrum from transparent linear models to opaque gradient-boosted ensembles and inherently inter-

pretable Generalized Additive Models (GAMs), the study exposes the practical trade-off between predictive accuracy and structural interpretability in a realistic large-scale regression setting (Molnar, 2022). An additional motivation for the multi-model design is the interactive comparison application developed as part of this project: the third page of the accompanying Streamlit dashboard allows users to provide identical apartment-level inputs and observe how predictions and their feature-level decompositions differ across model families, making the accuracy-interpretability trade-off directly tangible. Table 4.1 summarises the model portfolio.

Two candidate models were initially considered but excluded from the main evaluation. A *Decision Tree* regressor was tested in preliminary experiments but excluded from the main model portfolio because it did not add analytical value beyond the selected baseline and ensemble/interpretable model set under the final evaluation protocol. *TabPFN* (Hollmann et al., 2025), a transformer-based Prior-Fitted Network for small tabular datasets, was initially planned for inclusion but was superseded by *TabICLv2* (Qu et al., 2026) prior to training; as TabICLv2 offers broader scalability and native support for regression tasks, TabPFN is not evaluated in this study.

Table 4.1: Model portfolio overview. All models are applied identically to Model A (rent price, EUR/m^2) and Model B (log time-on-market). ✓ denotes inclusion in Bayesian hyperparameter search.

Model	Family	Interpretability	Native NaN	Tuned
Ridge Regression	Linear	Global (coefficients)	No	✓
XGBoost	Ensemble	Post-hoc via SHAP	Yes	✓
LightGBM	Ensemble	Post-hoc via SHAP	Yes	✓
EBM	GAM	Inherent (shape functions)	No	✓
IGANN	GAM	Inherent (shape functions)	No	✓
TabICLv2	TFM	No model-specific explanation	No	
TabPFN†	TFM	—	Yes	

† Planned but not evaluated; superseded by TabICLv2 prior to training.

For a detailed description of every used model look at B and C

4.3.1 Post-hoc Explainability: SHAP

To complement the inherent interpretability provided by EBM and IGANN, SHAP (SHapley Additive exPlanations) (Lundberg and Lee, 2017) is applied to the two gradient-boosted ensemble models as a post-hoc explainability layer. SHAP decomposes a model prediction for a given observation into additive feature contributions by computing Shapley values derived from cooperative game theory. For tree-based models, the TreeSHAP

algorithm computes exact Shapley values in polynomial time by exploiting the tree structure, making it computationally tractable at the present dataset scale.

Two levels of SHAP analysis are employed. At the global level, beeswarm plots of SHAP values aggregated over the full test set rank features by their average absolute contribution and reveal the direction and distribution of individual feature effects across observations; this serves as a basis for comparing the importance rankings of the gradient-boosted models against those of the GAMs in Section 5.4. At the local level, waterfall plots decompose individual listing predictions into their feature-level contributions, enabling an interpretable inspection of why a specific apartment receives a particular price or duration estimate. The local SHAP analysis is directly exposed in the third page of the Streamlit application, where predictions from all available models can be examined on identical input data.

The shape functions of EBM and IGANN represent globally learned marginal effects that are fixed structural properties of the fitted model, derived entirely from the training process. SHAP values, by contrast, are instance-specific and depend on the background distribution of the training data used to compute Shapley value expectations. The two approaches therefore provide complementary rather than equivalent perspectives on feature importance, and discrepancies between them constitute an informative diagnostic signal that is examined in Section 5.4.

4.4 Evaluation Methodology

4.4.1 Evaluation Metrics

Three complementary regression metrics are reported for all models and both targets. Mean Absolute Error (MAE) is the primary metric, selected for its robustness to the skewed distributions and outliers characteristic of rental price and listing duration data and for its direct interpretability in the unit of the target variable. Root Mean Squared Error (RMSE) is reported as a secondary metric; by squaring residuals before averaging, it penalises large individual errors more heavily than MAE and thereby highlights systematic model failures on atypical listings. The coefficient of determination R^2 expresses the share of target variance explained by the model and provides a scale-independent measure of relative fit.

For Model A, all metrics are expressed in the natural scale of the target variable (EUR/m^2). For Model B, the target variable is $\log(\text{laufzeittage} + 1)$; predictions and observed values are therefore back-transformed via $\exp(\hat{y}) - 1$ before MAE and RMSE

are computed, so that both error metrics are reported in days and results remain directly comparable across model families. It should be noted that R^2 for Model B is expected to be structurally lower than for Model A: because a substantial share of listings is rented within seven days, the target distribution exhibits a pronounced floor effect that limits the explanatory ceiling of any regression model.

4.4.2 Train/Test Split and Cross-Validation

The full processed dataset is divided into a training set (84.2 %) and a held-out test set (15.8 %) using a temporal split at listing year 2022: all observations with `ajahr` ≤ 2022 form the training partition (listing years 2007–2022; $n = 968{,}884$ in the 10 % sample); all observations with `ajahr` > 2022 form the held-out test partition (listing years 2023–2025; $n = 181{,}662$). A random split is deliberately avoided, as it would allow models to exploit future listing information during training and thereby produce overly optimistic performance estimates for a deployment scenario where only historical data is available at training time.

During hyperparameter search (Section 4.5), model performance is estimated via 3-fold cross-validation applied to a stratified 1 % subsample of the training set (approximately 117 000 rows). This configuration represents a deliberate trade-off between statistical robustness and computational feasibility given the time budget constraints of the study. After hyperparameter search, each model is retrained once on the full training set using the best-found configuration and evaluated a single time on the held-out test set; the resulting MAE, RMSE, and R^2 values constitute the final reported metrics in Section 5.2 and Section 5.3.

4.5 Hyperparameter Tuning

Hyperparameter optimization is conducted using the *Optuna* framework (Akiba et al., 2019) with the Tree-structured Parzen Estimator (TPE) as the acquisition function. TPE is a sequential model-based Bayesian optimization strategy that builds separate non-parametric density estimates for promising and unpromising hyperparameter configurations and proposes the next trial by maximizing the expected improvement ratio. Compared to grid or random search, Bayesian optimization concentrates the trial budget in regions of the hyperparameter space that have empirically performed well in preceding trials.

A separate Optuna study is created for each combination of model family and target variable, yielding ten independent optimization runs in total. All studies use the same

objective: minimize the mean MAE across 3-fold cross-validation folds on the 1 % stratified subsample described in Section 4.4.2. Studies for LightGBM, XGBoost, EBM, and IGANN are initialized with a warm-start trial that enqueues a manually configured parameter set derived from prior experiments; this ensures that the first evaluated point is already a reasonable solution and that subsequent TPE proposals can build on meaningful performance signal immediately. Table 4.2 summarises the per-study timeout budget. The substantially larger budget for EBM reflects its single-threaded training procedure, which results in considerably longer per-trial wall-clock times relative to the parallelized gradient-boosted tree implementations.

Table 4.2: Bayesian hyperparameter optimization budget per Optuna study. Each study runs independently for Model A and Model B.

Model	**Tuned Parameters**	**Timeout per Target**
Ridge Regression	1	2 min
LightGBM	8	30 min
XGBoost	8	30 min
IGANN	5	200 min
EBM	6	200 min
Total (both targets)		$\approx$ 15.4 h

The search spaces for each model family are listed in Table 4.3. Log-uniform sampling is applied to parameters that span several orders of magnitude (learning rates, regularization strengths), while integer and uniform continuous distributions are used for bounded parameters. For IGANN, the lower bound of `elm_alpha` is set to 10^{-3}; preliminary experiments demonstrated that smaller values produce near-singular ELM matrices and cause numerical failures during cross-validation, as discussed in Section C.

Table 4.3: Hyperparameter search spaces. Log: log-uniform sampling; Int: integer uniform; Cont: continuous uniform.

Model	Hyperparameter	Range	Sampling
Ridge	`alpha`	$[10^{-4}, 10^{4}]$	Log
LightGBM	`n_estimators`	[100, 1500]	Int
	`learning_rate`	$[10^{-3}, 0.3]$	Log
	`max_depth`	[3, 12]	Int
	`num_leaves`	[20, 300]	Int
	`subsample`	[0.5, 1.0]	Cont
	`colsample_bytree`	[0.5, 1.0]	Cont
	`reg_alpha`	$[10^{-8}, 10]$	Log
	`reg_lambda`	$[10^{-8}, 10]$	Log
XGBoost	`n_estimators`	[100, 1500]	Int
	`learning_rate`	$[10^{-3}, 0.3]$	Log
	`max_depth`	[3, 10]	Int
	`subsample`	[0.5, 1.0]	Cont
	`colsample_bytree`	[0.5, 1.0]	Cont
	`min_child_weight`	[1, 10]	Int
	`reg_alpha`	$[10^{-8}, 10]$	Log
	`reg_lambda`	$[10^{-8}, 10]$	Log
EBM	`max_bins`	[64, 512]	Int
	`max_interaction_bins`	[8, 64]	Int
	`interactions`	[0, 20]	Int
	`learning_rate`	$[5\times10^{-3}, 0.15]$	Log
	`min_samples_leaf`	[2, 20]	Int
	`max_rounds`	[1000, 5000]	Int
IGANN	`n_hid`	$\{64, \ldots, 256\}$	Int
	`n_estimators`	$\{500, \ldots, 1500\}$	Int
	`boost_rate`	[0.01, 0.5]	Log
	`elm_scale`	[0.1, 5.0]	Log
	`elm_alpha`	$[10^{-3}, 1.0]$	Log

4.6 Shape Function Analysis Methodology

The evaluation metrics of previous Section quantify how accurately each model predicts, but not what each model has learned about the individual features. The third goal of this work, introduced in Section 1.2, therefore turns from predictive accuracy to a structural comparison of the feature effects the models actually estimate. This section defines the object of that comparison and the protocol along which it is carried out. The unit of analysis throughout is the univariate marginal effect of a single feature x_j on the

prediction: for the additive glassbox models this is the shape function f_j of the additive structure itself, whereas for the gradient-boosted ensembles it is the closest post-hoc analogue obtainable from their SHAP attributions. The comparison runs along the two axes named in Section 1.2 - across the two target variables (identical feature set, different targets) and across the model types (intrinsic explanations versus post-hoc attributions).

Extraction of intrinsic shape functions. For the EBM, each f_j is read directly from the fitted model through the `explain_global` interface of the `interpret` library; the functions are piecewise-linear step functions defined on the histogram bins learned during training. For IGANN, each f_j is a smooth curve produced by the boosted ELM units and is materialised by evaluating the fitted model on a dense grid spanning the feature's observed range. In both cases the shape functions are mean-centred by construction, which fixes the additive identifiability of Equation (C.1): the global intercept β_0 absorbs the overall level, so that each f_j expresses a pure deviation from the mean prediction and the curves of different models become directly superimposable. Interpretation is correspondingly limited in the sparsely populated tails of a feature's empirical support, where few observations constrain the fit.

Post-hoc analogue for the black-box models. XGBoost and LightGBM possess no intrinsic per-feature function, because their effects are distributed across a large ensemble of interacting trees. The comparable object is therefore reconstructed from their TreeSHAP attributions (Lundberg, Erion, et al., 2020): plotting the SHAP value of a feature against its input value across the test set produces a representation that visually resembles a shape function. The essential difference is that this representation is a point cloud rather than a single curve. The vertical dispersion of the cloud at a fixed feature value is itself informative - for a purely additive main effect it would collapse to a line, so any spread quantifies the degree to which the feature interacts with the remaining features. The central tendency of the cloud is assessed visually rather than through a fitted trend line, and this visual read serves as the point of comparison with the GAM shape functions.

Comparability and comparison criteria. The two targets are differently scaled - Target A in euros per square metre, Target B on the $\log(1 + x)$ scale of the listing duration - so the numerical magnitude of an effect is not comparable across targets. Each shape function is kept in the native, mean-centred scale of its own target and the curves are placed side by side rather than forced onto a shared axis. The comparison itself proceeds not on magnitude but on the structural properties of each curve: the sign of the effect, its monotonicity and its functional form (linear, saturating, u-shaped).

The two comparison axes. Along the first axis, the mean-centred effects $f_j^{(A)}$ and $f_j^{(B)}$ of each analysed feature are compared and classified as *concordant* (the feature moves both targets in the same direction), *opposing* (it moves them in opposite directions), *mixed* (the direction reverses within the feature's own range), or *inconclusive* (cross-method disagreement or an overall weak, noisy signal precludes a reliable cross-target label). This classification is the direct empirical test of the premise that rental price and time-on-market rest on partly distinct feature structures despite the two targets being only weakly correlated at the aggregate level (cf. Section 4.1). Along the second axis, the intrinsic shape function is compared, per target, against the post-hoc SHAP pattern. When both approaches agree in sign and form, the observed effect can be considered a robust property of the data rather than an artefact of one particular model family. Where they disagree, the divergence is not dismissed as noise but traced to a specific cause: either to an interaction effect, visible through the SHAP dispersion, instability in sparsely populated regions of the shape function; or ambiguous attribution under strong feature correlation, as is expected for mutually dependent features such as a flat's floor level and total number of floors (Molnar, 2022).

5 Results

5.1 Hyperparameter Tuning Results

Table 5.1 reports the best hyperparameter configurations identified by the Bayesian optimization procedure described in Section 4.5. All values represent the parameter combination that minimized the 3-fold cross-validation MAE on the stratified 1 % subsample for the respective target variable. Because the optimization sample constitutes only a fraction of the full training set, and because gradient-boosted ensemble methods in particular benefit substantially from larger training volumes, the cross-validation MAE estimates obtained during hyperparameter search are expected to exceed the final test-set errors reported in Sections 5.2 and 5.3. The hyperparameter configurations should therefore be interpreted as well-calibrated starting points rather than as direct performance predictors.

It should further be noted that all cross-validation MAE values produced during optimization for Model B (time-on-market) are expressed in log-units of the target variable $\log(\text{laufzeittage} + 1)$ and are not directly comparable to the back-transformed day-scale metrics reported in Section 5.3.

Table 5.1: Best hyperparameter configurations identified by Bayesian optimization (Optuna TPE, 3-fold CV, 1 % stratified sample). Model A: rent price (EUR/m^2); Model B: log time-on-market. Trials: number of completed Optuna trials within the timeout budget (not recorded for Ridge).

Model	**Hyperparameter**	**Model A**	**Model B**	**Trials (A / B)**
LightGBM	`n_estimators`	1 499	1 116	151 / 205
	`learning_rate`	0.0264	0.0059	
	`max_depth`	12	12	
	`num_leaves`	244	285	
	`subsample`	0.831	0.711	
	`colsample_bytree`	0.536	0.508	
	`reg_alpha`	0.083	$<10^{-6}$	
	`reg_lambda`	0.580	$<10^{-3}$	
XGBoost	`n_estimators`	1 427	1 304	110 / 124
	`learning_rate`	0.0362	0.0062	
	`max_depth`	8	10	
	`subsample`	0.939	0.600	
	`colsample_bytree`	0.514	0.548	
	`min_child_weight`	3	2	
	`reg_alpha`	$<10^{-5}$	4.718	
	`reg_lambda`	$<10^{-5}$	0.404	
EBM	`max_bins`	359	507	14 / 17
	`max_interaction_bins`	38	43	
	`interactions`	20	8	
	`learning_rate`	0.0881	0.0051	
	`min_samples_leaf`	16	15	
	`max_rounds`	4 996	3 943	
IGANN	`n_hid`	64	64	4 / 6
	`n_estimators`	500	500	
	`boost_rate`	0.19	0.19	
	`elm_scale`	0.772	0.772	
	`elm_alpha`	0.133	0.133	
Ridge	`alpha`	77.3	9 992	—

Several observations emerge from the configurations in Table 5.1. *LightGBM* selects a large number of estimators in both targets, near the upper bound of the search range, paired with a comparatively low learning rate; this is consistent with the well-established principle that shallow learning steps generalize better than aggressive single-step updates (Chen and Guestrin, 2016). The systematically lower learning rates chosen for Model B relative to Model A indicate that the duration signal is more difficult to fit and requires finer incremental adjustments. *XGBoost* exhibits the same high-estimator, low-learning-rate pattern; in addition, the regularization parameters for Model B are substantially larger (`reg_alpha` = 4.72, `reg_lambda` = 0.40) compared to Model A (both near zero), suggesting a higher overfitting risk on the duration target that the optimizer mitigates through explicit weight penalization. The *EBM* configuration for Model A exhausts the permitted pairwise interaction budget (`interactions` = 20), indicating that interaction terms carry genuine explanatory value for price prediction; for Model B, only 8 interaction pairs are retained, pointing to a more additively structured relationship between apartment features and time-on-market. For *IGANN*, the optimization run yielded identical best configurations for Model A and Model B (Table 5.1), with comparatively few completed trials within the timeout budget. These results should therefore be interpreted as a stable local optimum under the given search setup rather than as evidence of a strongly target-specific global optimum for IGANN hyperparameters. The *Ridge* regularization parameter differs by more than two orders of magnitude between targets ($\alpha = 77.3$ for Model A versus $\alpha = 9{,}992$ for Model B), reflecting the substantially weaker linear signal present in the duration target; the Model B optimum lies at the upper edge of the search space $[10^{-4}, 10^{4}]$, suggesting that even stronger shrinkage might have been selected had the range permitted it.

TabICLv2 was not subjected to hyperparameter tuning for the reasons set out in Section 4.3: as a pretrained in-context learner, TabICLv2 has no conventional training hyperparameters amenable to gradient-based or Bayesian optimization, and data-centric tuning would compromise the uniform feature matrix applied to all other models.

Figure A.1 illustrates the progression of the Optuna optimization across trials for each model. The results indicate that IGANN and EBM benefited substantially from hyperparameter tuning, whereas LightGBM and XGBoost already exhibited strong baseline performance and consequently showed only modest improvements during the optimization process. Furthermore, Figure A.2 presents the Spearman rank correlation based hyperparameter importance analysis, computed exemplarily for the price prediction task for each model.

5.2 Model Performance: Rent Price (Model A)

All six models were trained on the temporal training partition covering listing years 2007–2022 (968,884 observations; 84.2 % of the full 10 % sample; 32 features) and evaluated on the held-out test partition covering listing years 2023–2025 (181,662 observations; 15.8 %). The temporal cutoff ensures that no future market information enters model training, establishing a deployment-realistic benchmark at the cost of more conservative performance estimates compared to a naive random split.

Table 5.2 reports MAE, RMSE, and R^2 on the test set for all six models, ranked by MAE.

Table 5.2: Test-set performance for Model A (rent price prediction, EUR/m^2). Training set: listing years 2007–2022 (n = 968,884); test set: listing years 2023–2025 (n = 181,662). All metrics expressed on the original EUR/m^2 scale. †TabICLv2 evaluated in zero-shot mode without target-specific hyperparameter tuning (see Section 4.3).

Model	**MAE** (EUR/m^2)	**RMSE** (EUR/m^2)	R^2
LightGBM (tuned)	2.005	2.930	0.706
XGBoost (tuned)	2.024	2.959	0.701
EBM (tuned)	2.441	3.469	0.589
TabICLv2†	2.791	3.882	0.485
IGANN (tuned)	2.837	3.858	0.491
Ridge (tuned)	2.889	4.171	0.405

Black-box performance ceiling. LightGBM and XGBoost occupy the two top positions with near-identical metrics (MAE difference: 0.019 EUR/m^2; R^2 difference: 0.006), confirming the well-established strength of gradient-boosted ensembles on tabular data (Chen and Guestrin, 2016). Both models explain approximately 70 % of the variance in out-of-sample rent prices with a mean absolute prediction error of roughly 2 EUR/m^2.

Interpretability–accuracy trade-off. The EBM closes the interpretability gap at a moderate accuracy cost: its MAE of 2.441 EUR/m^2 lies 0.44 EUR/m^2 (+22 %) above LightGBM, while its $R^2 = 0.589$ remains above 0.5. Crucially, the EBM produces fully auditable, shape functions for every feature and pairwise interaction (see Section 5.4), which constitutes the primary scientific contribution of this study. The EBM therefore represents a viable interpretable alternative to gradient-boosted ensembles for hedonic rent price modelling.

IGANN achieves $R^2 = 0.491$ and places fifth despite hyperparameter tuning, ranking below the zero-shot TabICLv2 in MAE (2.837 versus 2.791 EUR/m^2). The zero-shot performance of TabICLv2 illustrates the generalization capacity of tabular foundation

models pretrained on large external corpora; its black-box architecture, however, precludes feature-level interpretation. Ridge regression, as the linear baseline, yields the weakest results ($R^2 = 0.405$), confirming that non-linear feature relationships carry substantial explanatory power in rent price prediction.

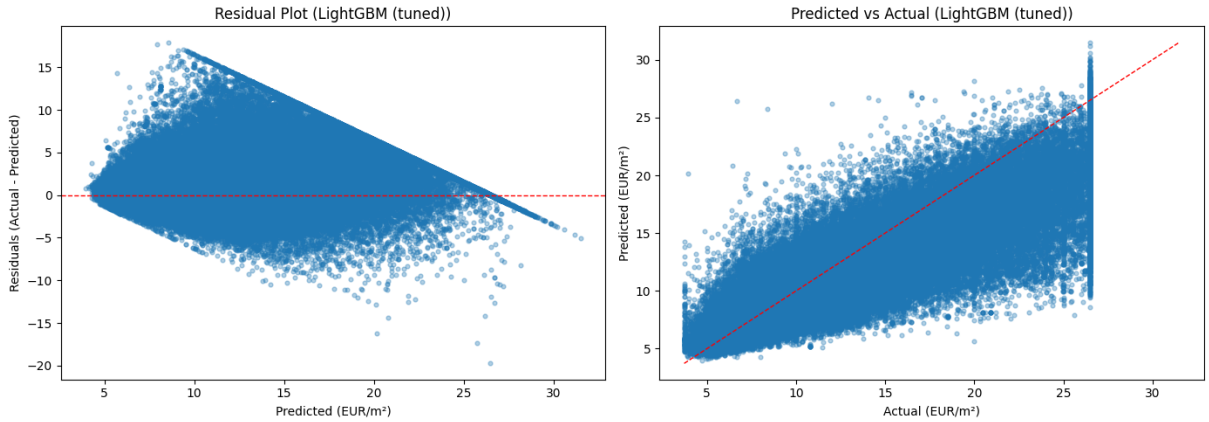


Figure 5.1: Residual plot (left) and predicted-versus-actual scatter (right) for LightGBM (best model) on the Model A test set (listing years 2023–2025). Target unit: EUR/m^2. The red dashed line marks zero residuals (left) and the identity line (right).

Figure 5.1 shows the residual diagnostic for LightGBM. Residuals are centred near zero across most of the prediction range, with systematic underprediction for high-rent listings exceeding 15 EUR/m^2 – a characteristic pattern in right-skewed real estate distributions where extreme prices are underrepresented in the training data.

5.3 Model Performance: Time-on-Market (Model B)

Model B uses the same temporal training and test partitions as Model A. The target variable is log(laufzeittage + 1); predictions are back-transformed via $\exp(\hat{y}) - 1$ and MAE and RMSE are computed on the day scale to enable direct interpretation, while R^2 is computed on the log-scale target as specified in Section 4.4.1.

Table 5.3: Test-set performance for Model B (time-on-market prediction). Training set: 2007–2022 (n = 968,884); test set: 2023–2025 (n = 181,662). MAE and RMSE computed on back-transformed predictions ($\exp(\hat{y}) - 1$) and expressed in days; R^2 computed on the log-scale target. †TabICLv2 evaluated in zero-shot mode. ‡R^2 recomputed on the original day scale for the best model only: 0.048.

Model	**MAE** (days)	**RMSE** (days)	R^2 (log scale)
XGBoost (tuned)‡	14.35	25.67	0.086
LightGBM (tuned)	14.37	25.71	0.082
Ridge (tuned)	14.57	27.15	0.054
EBM (tuned)	14.65	26.14	0.049
TabICLv2†	14.67	25.89	0.015
IGANN (tuned)	16.42	26.29	−0.164

Low overall explanatory power. All models achieve R^2 values between −0.164 and 0.086 on the log-scale target, confirming the structural difficulty anticipated in Section 4.4.1. This pattern is consistent with findings from the time-on-market literature reviewed in Section 3.4: listing duration is substantially determined by factors not recorded in the advertisement data, including landlord response speed, scheduling logistics, pricing strategy, and timing-driven demand fluctuations (cf. Haurin, 1988; Anglin, Rutherford, and Springer, 2003). The floor effect – a substantial share of listings rented within seven days – additionally suppresses the theoretical R^2 ceiling regardless of model complexity.

Model convergence and comparison to a naive baseline. Excluding IGANN, the five remaining models converge to a narrow MAE band of 14.35–14.67 days. A naive constant-median baseline – always predicting the training-set median listing duration of 15 days – achieves a test-set MAE of approximately 14.97 days. The best model (XGBoost) thus improves upon this baseline by only 0.62 days ($\approx 4\,\%$), indicating that the observable listing features carry very limited signal for duration prediction once the overall distributional level is accounted for.

Post-2022 distributional shift. The test partition covers listing years 2023–2025, a period following the 2022 interest-rate shock that substantially altered rental market dynamics in Germany. Training data drawn from 2007–2022 therefore represents a market regime that differs from the test period in demand intensity, listing durations, and tenant behaviour. This distributional shift between training and test regime constitutes an additional source of prediction error that cannot be mitigated by further model tuning on the available features.

IGANN failure under weak signal. IGANN achieves a negative R^2 of −0.164, indicating predictions less accurate than a constant mean predictor. This outcome is

attributable to the architectural constraints of the IGANN framework: the single-hidden-layer Extreme Learning Machine boosting structure provides limited capacity for modelling weak, noisy regression targets (cf. Kraus et al., 2024). When the target signal is thin – as quantified by the narrow inter-model MAE spread – the ELM output layer amplifies residual noise rather than capturing the underlying pattern. EBM avoids this failure mode ($R^2 = 0.049 > 0$), albeit with similarly modest explanatory power.

Note on R^2 scale reporting. R^2 is not invariant under non-linear transformations and cannot be converted between the log scale and the day scale analytically; it must be recomputed on the desired scale. For XGBoost, the day-scale R^2 equals 0.048, lower than the log-scale value of 0.086, because the right-skewed distribution of listing durations (test-set median: 12 days; 90th percentile: 48 days) causes a small number of long-running listings to dominate squared-error terms on the original scale. The log-scale R^2 is therefore the more informative summary statistic and is reported uniformly for all models in Table 5.3.

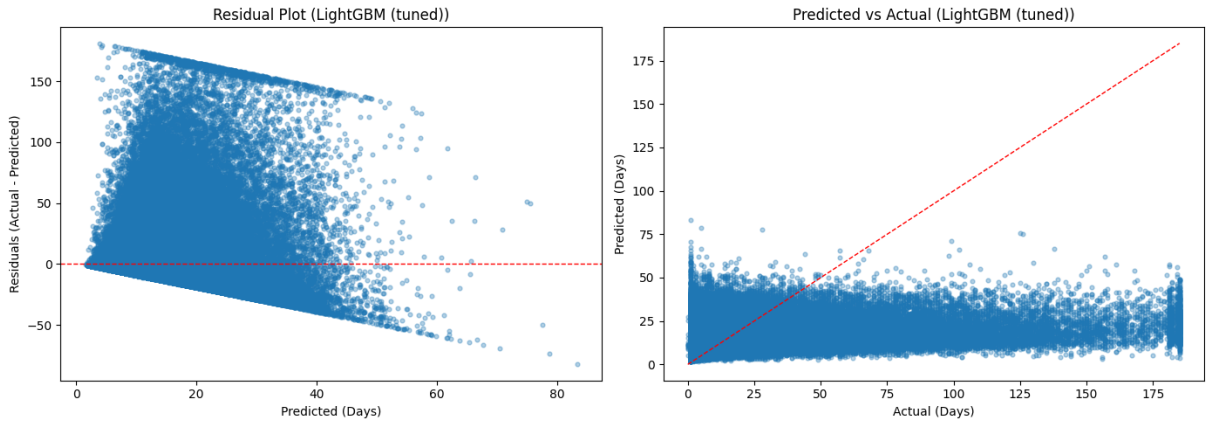


Figure 5.2: Residual plot (left) and predicted-versus-actual scatter (right) for LightGBM on the Model B test set (listing years 2023–2025). Both axes expressed in days after back-transformation.

Figure 5.2 shows the residual diagnostic for LightGBM. The model's predictions cluster near the central tendency, producing systematic underprediction for listings with durations exceeding 30 days – a direct consequence of the weak signal and the pronounced floor effect in the target distribution.

5.4 Shape Function Comparison: Price vs. Time-on-Market

Sections B and C established EBM and XGBoost as the glassbox and black-box representatives, respectively: IGANN's negative R^2 on Model B makes its duration shapes unreliable, while XGBoost and LightGBM importances agree closely enough ($\rho = 0.835, 0.986$; weighted $\tau = 0.844, 0.942$; see Appendix) to treat XGBoost as representative of the black-box class. The comparison below follows the two-step framework of Section 4.6: EBM's intrinsic shape functions are compared against XGBoost's TreeSHAP dependence curves *within* each target (cross-method), and *between* price and duration (cross-target). Each feature is labelled `agree`/`partial`/`disagree` on the first axis and `concordant`/`opposing`/`mixed`/`inconclusive` on the second; the full classification is given in Table E.1.

Table 5.4: Cross-method agreement between EBM and XGBoost (32 features).

Label	Features	Share
`agree`	18	56.2 %
`partial`	11	34.4 %
`disagree`	3	9.4 %

Cross-Method Robustness EBM and XGBoost agree on both shape and direction for a majority of features (56.2 %), and for a further 34.4 % the two methods agree in the densely populated core of the range, diverging only in sparse tails or on one of the two targets. The two methods disagree outright on only three features, all of them nominal categoricals, possibly reflecting inconsistent categorical encodings between the two pipelines.

A second, more informative source of `partial` divergence involves features with strong empirical correlation to other predictors (e.g. `anzahletagen` and `etage`): EBM concentrates shared signal from correlated predictors in a single main effect, whereas TreeSHAP distributes the attribution across all of them. This is the same attribution asymmetry underlying the comparatively low importance-rank correlation reported earlier for Model A ($\rho = 0.637$) relative to Model B ($\rho = 0.882$), whose leading predictors are largely uncorrelated. Where this divergence appears, the cross-target label is reported as `inconclusive` rather than forced into a substantive category.

Table 5.5: Cross-target classification of feature effects (32 features).

Label	Features	Share
`opposing`	14	43.8 %
`concordant`	8	25.0 %
`inconclusive`	7	21.9 %
`mixed`	3	9.4 %

Cross-Target Patterns The dominance of the `opposing` class is the central empirical result of this section, but its interpretation requires care. `Opposing` means only that price and duration move in opposite directions for a given feature; it does not by itself imply a single mechanism. Two distinct patterns underlie it. For most amenity and proximity features (`einbaukueche`, `balkon`, `dist_supermarket`), a feature that raises the price also shortens the duration – consistent with a simple desirability account: more attractive units command a premium and let faster. For a second, smaller group like `foerderung`, the feature lowers the price and lengthens the duration instead. Nor is this confined to a few exceptions: within the location theme, where five of six features are classified as opposing (Section 5.4), even the direction of the underlying effect is not consistent. The opposing relationship is therefore a robust structural regularity in this market, but it reflects more than one mechanism and should not be read as evidence for a single latent desirability dimension.

The concordant minority (25.0 %) is the more informative group precisely because desirability alone cannot explain it. Two qualifications apply. For `aufzug`, `gaestewc` and `hat_parkplatz`, the duration effect is negligible - these are effectively price-only effects whose concordance is an artefact of a near-zero second effect rather than a substantive finding. The remaining, substantively concordant cases (e.g. `alter_bauwerk` and the upper tail of `wohnflaeche`) describe niche market segments - characteristics that justify a price premium precisely because they serve a small, slow-moving demand segment. Segmentation, not desirability, is the operative mechanism here.

Duration effects are also systematically small. Excepting `freiab_monate` (+0.75 log-days, roughly a doubling of listing duration), no feature moves duration by more than $\approx \pm 0.2$ log-days ($\approx 20\,\%$), consistent with Model B's weak explanatory power ($R^2 = 0.049$–0.086, Section 5.3): the duration signal is real but small relative to unobserved determinants of time-on-market.

Table 5.6: Cross-method agreement and dominant cross-target class by feature theme.

Theme	n	`agree+partial` rate	Dominant cross-target class
Amenities	8	100.0 %	`opposing` (50.0 %)
Building	3	100.0 %	`concordant/opposing/mixed` (33.3 % each)
Dwelling	7	85.8 %	`inconclusive` (42.9 %)
Energy	3	33.3 %	`inconclusive` (66.7 %)
Location	6	100.0 %	`opposing` (83.3 %)
Meta	5	100.0 %	`opposing` (60.0 %)

Thematic Structure Two patterns in Table 5.6 deserve comment beyond the raw shares. Dwelling's high inconclusive rate (42.9 %) is not a uniform failure: for `badezimmer` and `etage`, both methods agree on the price effect and only duration is disputed, echoing the wider Model A/Model B gap in explanatory power (Section 5.3) within a single theme; the third case, `kategorie_Wohnung`, is nominal and inconclusive for the same reason as the Energy features below. Meta's aggregate figure (60 % opposing) likewise compresses considerable heterogeneity: it spans both the dataset's strongest duration driver (`freiab_monate`, discussed below) and features with much weaker or counterintuitive effects, so the theme-level share should not be read as describing a single uniform mechanism.

Energy and Amenities require no further comment: Energy's low robustness (33.3 %) reflects the encoding concern raised in Section 5.4 rather than a new finding, and Amenities' concordant cases are already explained by negligible duration effects (Section 5.4). Building, with only three features, is too small to generalise beyond the individual cases discussed above. Location stands out as the one theme where high method-level robustness (100 %) translates into an equally clear substantive picture: five of its six features are opposing, with only `dist_transit` too noisy to classify.

Detailed Feature Analysis Four features are examined in detail, selected to cover the main themes and to include both the strongest and the most instructive cases.

`wohnflaeche` (Figure 5.3). EBM and XGBoost trace near-identical curves for both targets, yet the price-duration relationship reverses across the size range (`mixed`), moving oppositely through most of the distribution before aligning at the top end. This points to three market segments rather than a single size gradient: scarce micro-apartments and large premium units both command a premium and let slowly, while mainstream mid-size units get cheaper without letting faster.

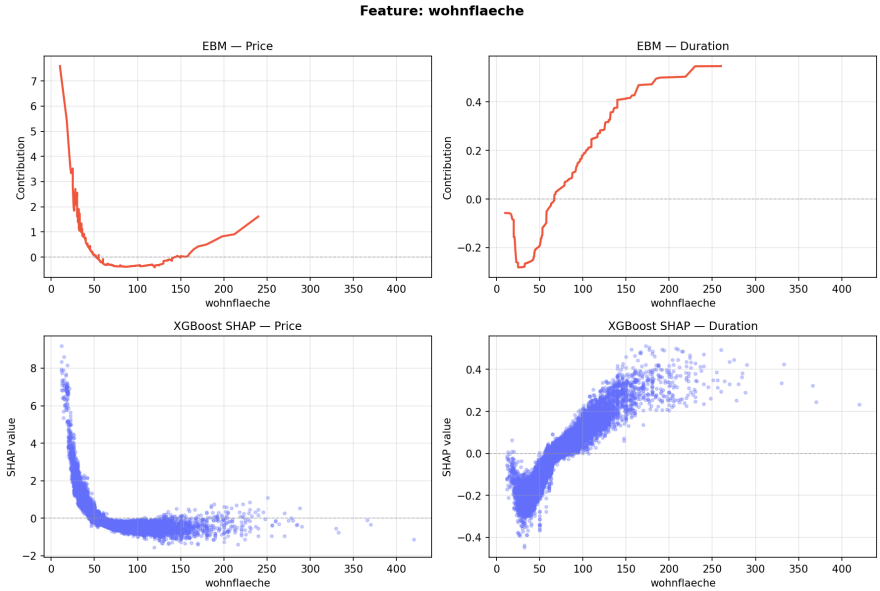


Figure 5.3: Shape functions for `wohnflaeche`: EBM (top) vs. XGBoost SHAP (bottom), price (left) vs. duration (right).

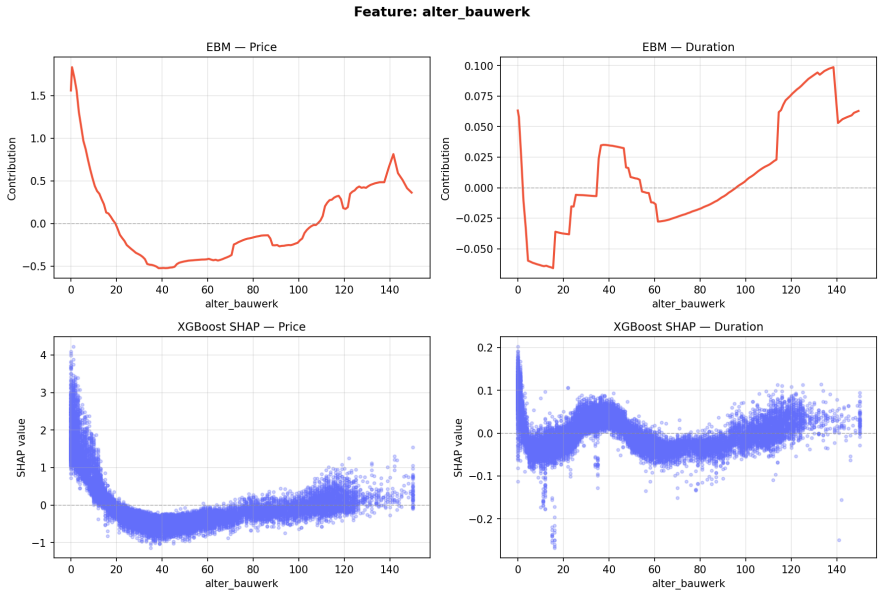


Figure 5.4: Shape functions for `alter_bauwerk`: EBM (top) vs. XGBoost SHAP (bottom), price (left) vs. duration (right).

`alter_bauwerk` (Figure 5.4). Both methods reproduce a U-shaped effect on both targets: new and old flats command a price premium and take longer to let, while mid-age stock is cheapest and fastest (`concordant`). This is a segment effect: niche building-age categories justify their premium through slower, more specific demand.

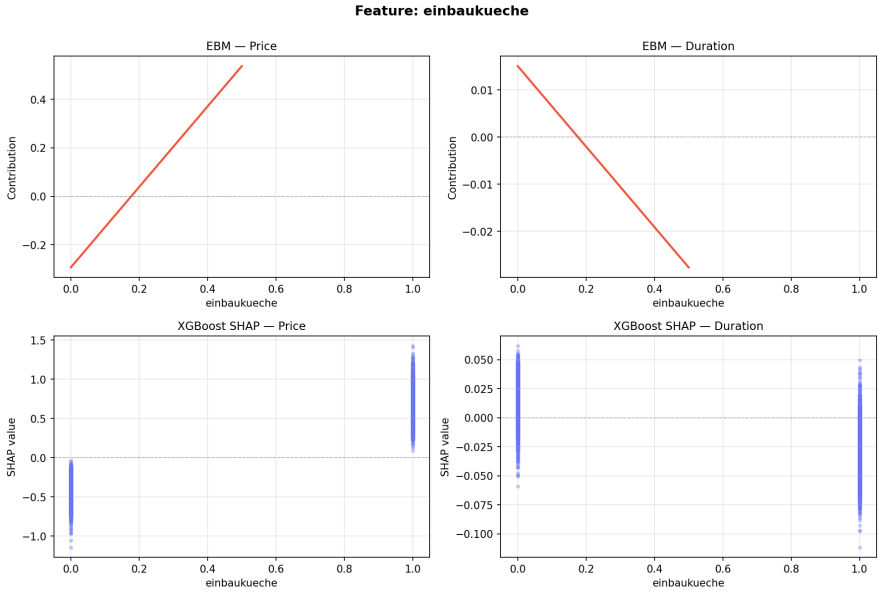


Figure 5.5: Shape functions for `einbaukueche`: EBM (top) vs. XGBoost SHAP (bottom), price (left) vs. duration (right).

`einbaukueche` (Figure 5.5). Near-perfect cross-method agreement on a simple step effect.[1] A fitted kitchen raises price and shortens duration (`opposing`) - the clearest desirability case in the dataset, with most of the value captured in price rather than speed of letting.

`dens_supermarket` (Figure 5.6). Both methods agree across the densely populated core of the range, with EBM becoming unstable only in the sparse tail. Price rises steeply before plateauing while duration falls monotonically (`opposing`) - consistent with retail density acting as a desirability proxy.

[1] For binary features, the EBM shape function is plotted at bin edges 0 and 0.5; the value at 0.5 corresponds to feature value 1.

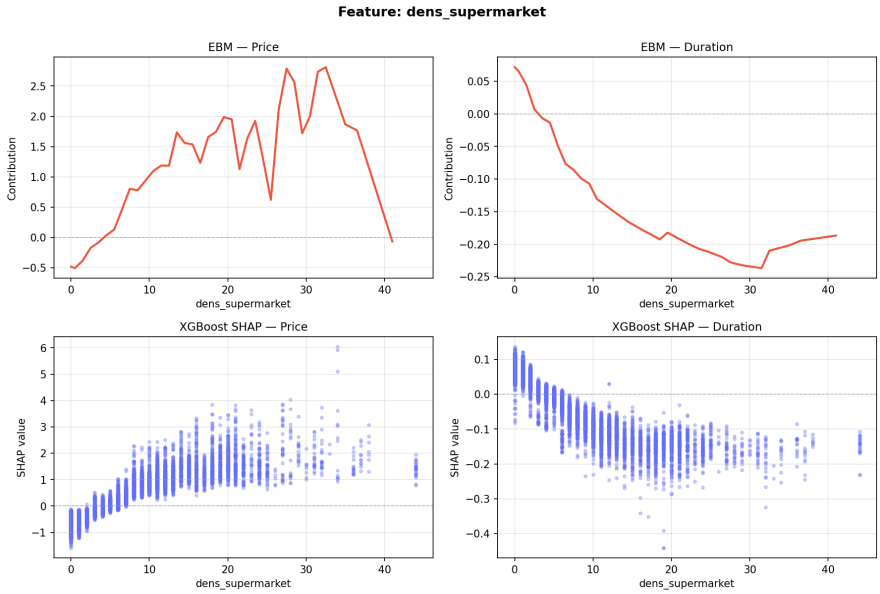


Figure 5.6: Shape functions for `dens_supermarket`: EBM (top) vs. XGBoost SHAP (bottom), price (left) vs. duration (right).

6 Discussion, Conclusion and Outlook

6.1 Summary of Explored Techniques

The experimental pipeline evolved through iterative refinement; several design choices were revised following empirical evaluation or computational constraints encountered during development. Table 6.1 documents all preprocessing variants, encoding strategies, sampling configurations, model families, and explainability methods considered throughout the study, alongside their inclusion status in the final evaluation pipeline. Two decisions had a material effect on the final results: the replacement of an initial random train/test split by the temporal partition described in Section 4.4.2, and the restriction of training to a 10 % stratified sample dictated by the memory and runtime constraints of TabICLv2 and IGANN.

Table 6.1: Techniques explored during pipeline development. ✓ denotes inclusion in the final evaluation pipeline; × denotes techniques considered but not carried into the final setup.

Category	Technique	Final	Note
Preprocessing	Median imputation (numerical)	✓	
	Mode imputation (categorical)	✓	
	Log-transform of target variable	✓	Applied to Model B; reduces right-skew of `laufzeittage`
	StandardScaler normalization	✓	Applied to Ridge and IGANN feature matrices
	Random train/test split	×	Initial approach; replaced by temporal split after identifying cross-year data leakage
Encoding	Ordinal encoding	✓	Quality and condition ordinals (`objektzustand`)
	One-hot encoding	✓	Nominal categoricals; Ridge, IGANN, TabICLv2
	Binary indicator flags	✓	Amenity features (`balkon`, `keller`, `aufzug`, `einbaukueche`)
	`erg_amd` as categorical	×	141 distinct categories; excluded from Ridge, IGANN, and TabICLv2 due to dimensionality
Sampling	Full dataset (∼11.5M rows)	×	Computationally infeasible for iterative model development and HPO
	1 % stratified subsample	✓	Bayesian HPO exclusively (≈117k training rows)
	10 % stratified sample	✓	Final training and held-out evaluation (n = 1,150,546)
Model families	Ridge Regression	✓	Linear baseline; interpretable via global coefficients
	Decision Tree	×	Preliminary experiments; no analytical value beyond selected model set
	XGBoost	✓	Black-box performance ceiling; post-hoc SHAP
	LightGBM	✓	Black-box performance ceiling; post-hoc SHAP
	EBM	✓	Glassbox; intrinsic shape functions for both targets
	IGANN	✓	Trained on both targets; Model B shape functions excluded ($R^2 = -0.164$)
	TabPFN	×	Superseded by TabICLv2 prior to training
	TabICLv2	✓	Zero-shot in-context learner; no target-specific tuning
Explainability	EBM intrinsic shape functions	✓	Primary glassbox analysis; both targets
	IGANN intrinsic shape curves	✓	Model A only; Model B excluded due to negative R^2
	TreeSHAP dependence plots	✓	Post-hoc analogue for XGBoost and LightGBM
	Ridge linear coefficients	✓	Diagnostic reference for linear feature effects

6.2 Interpretation of Results

Research question: divergent drivers. The central question posed in Section 2.3 asks whether rent price and time-on-market are governed by the same set of feature effects. The cross-target classification of Section 5.4 provides a direct empirical answer: 43.8 % of analysed features are classified as `opposing` (Table 5.5), making directional divergence the dominant structural regularity in this dataset. The weak marginal correlation between the two targets ($r = -0.119$, Section 4.1) is thus confirmed as the aggregate expression of widespread feature-level opposition rather than distributional noise.

Two distinct mechanisms underlie this regularity. The first is a desirability effect: amenity features such as `einbaukueche` and `balkon` raise price and simultaneously shorten duration, consistent with the interpretation that more attractive units command a premium and attract tenants faster. The second is a segmentation effect: niche characteristics, specifically building age extremes and the upper tail of `wohnflaeche`, justify a price premium precisely because they serve a small, slow-moving demand segment, producing concordant upward movement in both price and duration. The two mechanisms coexist and cannot be collapsed into a single latent desirability dimension. A structural anomaly reinforces this point: transit-stop density (`dens_transit`) runs counter to all other proximity features, suggesting that high transit density partly proxies for traffic-adjacent locations rather than acting as a clean amenity signal (Section 5.4).

Theoretical alignment. Model A results are consistent with the hedonic pricing framework of S. Rosen (1974): rental price decomposes into implicit marginal prices for individual dwelling attributes, and EBM's intrinsic shape functions make this decomposition directly observable without requiring post-hoc attribution. The substantially larger pairwise interaction budget selected during hyperparameter optimisation for Model A (20 terms) relative to Model B (8 terms, Table 5.1) further indicates that price formation is more interaction-driven than demand responsiveness, a structural asymmetry that additive post-hoc methods would conceal by distributing interaction signal across individual feature attributions.

For Model B, all models converge to $R^2 \leq 0.086$ (Section 5.3). This outcome aligns with search-theoretic accounts of time-on-market (Haurin, 1988; Anglin, Rutherford, and Springer, 2003): listing duration is primarily governed by unobserved matching dynamics, including landlord response behaviour, scheduling logistics, and tenant search intensity, none of which are encoded in the advertisement record. Structural dwelling features carry genuine but small directional signal for duration, as evidenced by the shape functions of Section 5.4; they do not, however, explain a substantial share of duration variance once

the overall distributional level is accounted for. Model B's near-baseline performance is therefore not a modelling failure but a theoretically coherent null result: the observable feature set is simply insufficient for duration prediction, and this limitation is intrinsic to the data.

Interpretable versus black-box models. For Model A, full interpretability is available at a moderate accuracy cost. EBM achieves $R^2 = 0.589$ and a test-set MAE of 2.44 EUR/m^2 against a black-box ceiling of $R^2 \approx 0.706$ and MAE $\approx$ 2.0 EUR/m^2 for LightGBM and XGBoost (Section 5.2). The resulting 22 % MAE premium purchases complete auditability: every feature's marginal effect is represented by a single, shape function that can be inspected, validated against domain knowledge, and communicated to non-technical stakeholders without requiring post-hoc approximations (Rudin, 2019). For applications requiring regulatory compliance or scientific reproducibility of feature effects, this trade-off is justified.

For Model B, the accuracy trade-off is immaterial because no model family achieves predictive performance beyond a naive baseline to a degree that would support deployment. The scientific contribution of interpretable models on this target is exclusively structural: EBM shape functions reveal the directional associations the model has extracted from the data, independent of whether that signal suffices for accurate prediction. This asymmetry is a substantive finding in its own right: the IML framework yields quantitative economic insight for price prediction and qualitative structural insight for duration prediction.

Cross-method robustness supports the validity of both conclusions. EBM and XGBoost agree on shape and direction for 90.6 % of features when `agree` and `partial` classifications are combined (Table 5.4), confirming that the reported feature effects reflect genuine data structure rather than artefacts of a particular model family (cf. Lundberg, Erion, et al., 2020; Molnar, 2022). Where the two methods diverge, the divergence traces to attribution asymmetry under feature correlation rather than to substantive disagreement about the underlying effect (Section 5.4).

6.3 Limitations and Outlook

6.3.1 Limitations

Platform sampling bias. The RWI-GEO-RED dataset is sourced exclusively from ImmoScout24, covering approximately 50 % of the German rental market. The platform overrepresents urban listings and digitally proficient landlords (cf. Heuer and Thiel, 2026). Rural and informal rental segments are systematically underrepresented. Shape

function profiles derived from this data may therefore overstate effects associated with urban amenities and cannot be assumed to generalise to non-platform market segments.

Limited spatial resolution. Exact address coordinates are not included in the Scientific Use File; location is approximated by the centroid of the 1×1 km INSPIRE grid cell (Section 4.1), which aggregates substantial within-cell heterogeneity. All engineered spatial features (POI distances and densities, `dist_to_center`) inherit this resolution limit, and micro-location attributes below the grid level, such as street-level characteristics or walkability scores, are established drivers of hedonic price variation (cf. Sirmans, Macpherson, and Zietz, 2005) that remain unobservable in this setting. Their absence is expected to reduce Model A's explanatory ceiling and to leave a portion of regional price variance unattributed.

Correlational shape functions. Shape functions quantify marginal predictive associations, not causal effects. No identification strategy is employed to separate the contribution of individual housing attributes from confounding by unobserved demand heterogeneity, landlord pricing behaviour, or regional market conditions. Causal claims are therefore outside the scope of this study.

Computational constraints. The scale of the RWI-GEO-RED dataset imposed binding resource constraints across the pipeline. Hyperparameter optimisation was confined to a 1 % stratified subsample with modest trial budgets: IGANN completed only 4 to 6 Optuna trials per target (Table 5.1), substantially fewer than required to characterise its hyperparameter surface reliably. Final model training was capped at a 10 % stratified sample. Gradient-boosted ensemble methods such as XGBoost and LightGBM scale favourably to larger training volumes and would have benefited from additional data; however, the memory and runtime constraints of TabICLv2 and IGANN imposed a common ceiling that was maintained across all models to preserve strict comparability.

Structural weakness of Model B. The low explanatory power of all time-on-market models ($R^2 \leq 0.086$) reflects an inherent limitation of the available feature set. Listing duration is substantially determined by unobserved factors such as landlord response behaviour, scheduling logistics, and pricing strategy that are not encoded in the advertisement data (cf. Haurin, 1988; Anglin, Rutherford, and Springer, 2003). The test partition (2023–2025) additionally falls after the 2022 interest rate shock, introducing a distributional shift between training and test regimes that further suppresses out-of-sample performance independently of model quality.

Pipeline implementation overhead. As the team's first large-scale tabular ML project, the study incurred non-trivial rework costs. An initial implementation applied a random rather than a temporal train/test split, which introduced data leakage across

listing years and invalidated early model evaluations. Identifying and correcting this error required substantial re-processing time that would otherwise have been available for extended hyperparameter search and additional modelling iterations.

6.3.2 Outlook

Spatial machine learning. The grid coordinates were used solely to compute distance and density features; the spatial dependence structure between neighbouring observations itself remains unmodelled. Incorporating spatial lag terms, district-level panel data, or Gaussian process priors for location effects would directly address the most consequential gap in the current feature matrix and is expected to raise the explanatory power of Model A substantially.

Text-based features. The full RWI-GEO-RED release includes free-text listing descriptions (`WM_SUF_mitText.csv`). Sentence-level embeddings derived from these descriptions could capture quality signals and presentation effects that structured fields do not encode, with potential gains for both predictive accuracy and the interpretability of amenity-related shape functions.

Cross-segment extension. The analysis is confined to the residential rental apartment segment (WM). Extending the two-target framework to house sales (HM) and condominium rentals (WK) within the same dataset would enable a systematic comparison of whether price formation and demand responsiveness mechanisms are segment-invariant or structurally distinct across housing types.

Causal identification. The Mietpreisbremse regulation (2015) and the Berliner Mietendeckel (2020–2021) provide quasi-experimental variation in price constraints that could support difference-in-differences or regression discontinuity identification strategies. Pairing such an approach with the interpretable ML framework developed here would allow shape functions to be reinterpreted as causal marginal effects rather than predictive associations, directly extending the scientific contribution of this study.

A Hyperparameter Tuning Plots

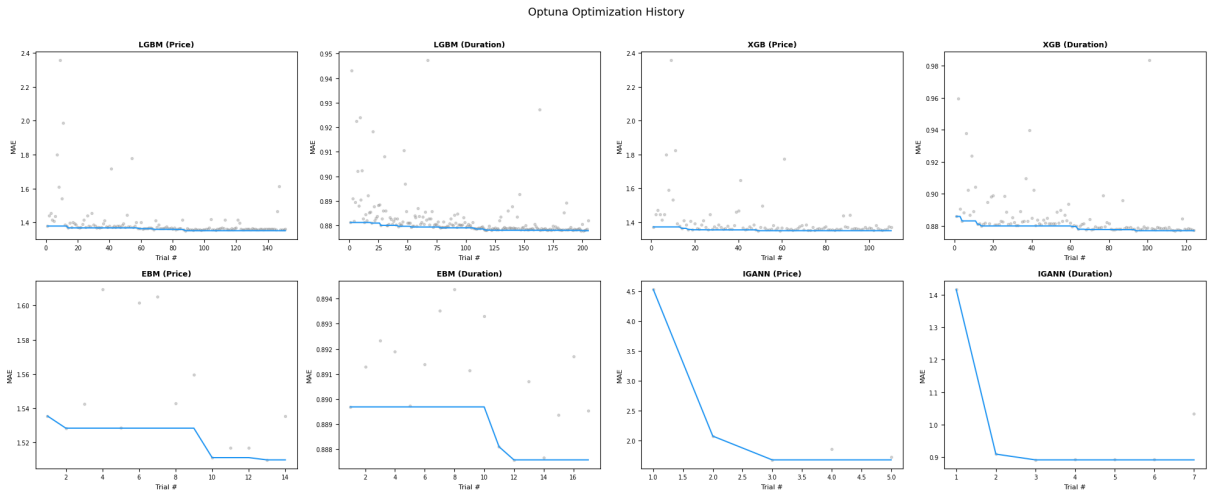


Figure A.1: Optimization history of the Optuna studies per model family and target.

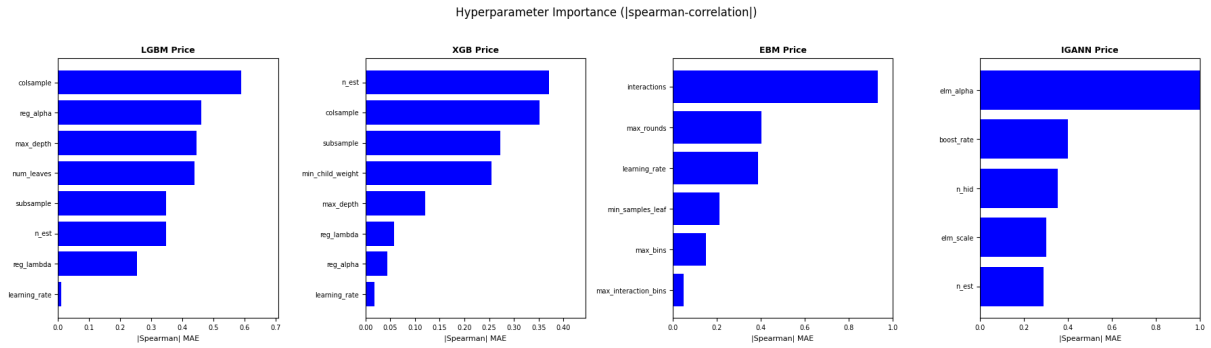


Figure A.2: Spearman rank correlation based hyperparameter importance per model family (Model A, rent price prediction).

B Baseline and Black-Box Models

Two model families serve distinct reference roles in the evaluation framework. The first family comprises interpretable baseline models whose purpose is to establish a lower bound on achievable predictive performance and to verify that the more complex model families deliver a meaningful gain in explanatory power. The second family comprises high-performance black-box models that define the empirical performance ceiling against which the interpretable models are ultimately judged.

Baselines. *Ridge Regression*, a regularized variant of ordinary least squares that penalizes the L_2 norm of the coefficient vector by a regularization strength α, serves as the primary linear baseline. Because the model assumes a strictly additive and globally linear relationship between each feature and the target variable, it is expected to underfit the complex non-linear and interaction-driven price structure of the German rental market. Its principal value in this study is therefore diagnostic rather than purely predictive: features whose Ridge coefficients are large in magnitude carry a meaningful linear signal and provide a theoretically grounded entry point before more flexible models are introduced. The regularization strength α was optimized via Bayesian hyperparameter search (see Section 4.5).

Gradient-Boosted Ensembles. *XGBoost* (Chen and Guestrin, 2016) and *LightGBM* (Ke et al., 2017) represent the performance ceiling tier. Both are gradient-boosted decision tree ensembles that iteratively fit shallow trees to the negative gradient of a differentiable loss function, but they differ in their tree-growth strategy. XGBoost grows trees level-wise using a histogram-based split-finding algorithm (`tree_method=hist`), constraining depth explicitly via the `max_depth` parameter, and is well-suited to CPU computation on large tabular datasets. LightGBM employs a leaf-wise growth strategy governed by `num_leaves`, which tends to produce deeper and more asymmetric trees, achieving lower training error per iteration at the cost of a higher overfitting risk on small samples. Both models handle missing values natively without requiring an imputation step, which is a practical advantage given the non-trivial missingness rates present in the feature matrix.

Their primary analytical role in this study is to serve as an accuracy benchmark and to provide SHAP-based post-hoc explanations as a contrast to the inherent shape functions of the GAM models.

Tabular Foundation Model (TabICLv2). *TabICLv2* (Qu et al., 2026) is a transformer-based in-context learner for tabular data that generates predictions by attending over the entire labelled training set at inference time, without fitting an explicit parametric function to the target dataset. This in-context mechanism constitutes a fundamentally different learning paradigm from the iterative optimization used by all other models in this study, and TabICLv2 is therefore included as an exploratory reference point representing the emerging class of tabular foundation models. Because no training-time parameter optimization occurs, TabICLv2 is not amenable to Bayesian hyperparameter search; the only alternative tuning strategy would be data-centric feature selection, which would violate the uniform feature matrix applied identically across all other models and was therefore not pursued. In the final experiment setup, TabICLv2 was run as a reference model without target-specific hyperparameter tuning, using a fixed-size random subset of the training data as in-context memory under hardware memory constraints. Its predictions were then evaluated once on the same held-out temporal test set (2023–2025) used for the other model families.

C Inherently Interpretable Models: EBM and IGANN

The scientific core of the methodological framework consists of two inherently interpretable Generalized Additive Models. Unlike post-hoc explanation methods, which produce approximations of a black-box model's behavior after the fact, inherently interpretable models are globally transparent by construction: their prediction function is fully decomposable into individual feature contributions that can be inspected, visualised, and compared without approximation (Molnar, 2022). Both models conform to the following additive prediction structure:

$$\hat{y} \;=\; \beta_0 \;+\; \sum_{j=1}^{p} f_j(x_j), \tag{C.1}$$

where β_0 is the global intercept, $p = 32$ denotes the number of input features, and $f_j \colon \mathbb{R} \to \mathbb{R}$ is a univariate shape function capturing the marginal effect of feature x_j on the prediction across its full observed value range. This additive structure enables the direct comparison of learned feature effects between Model A and Model B in Section 5.4: because $f_j^{(A)}(x_j)$ and $f_j^{(B)}(x_j)$ are both mean-centered marginal effects expressed in the scale of the respective target variable, they can be placed side-by-side to reveal whether a given feature drives rental price and time-on-market in the same or in opposing directions.

Explainable Boosting Machine (EBM). The *EBM* (Lou et al., 2013), implemented via the `interpret` library, extends the standard GAM formulation by incorporating automatically detected pairwise interaction terms:

$$\hat{y} \;=\; \beta_0 \;+\; \sum_{j=1}^{p} f_j(x_j) \;+\; \sum_{j<k} f_{jk}(x_j,\, x_k). \tag{C.2}$$

Each univariate shape function f_j is learned through a cyclic gradient-boosting procedure that fits shallow regression trees exclusively to one feature at a time, thereby preventing any given feature from absorbing variance that properly belongs to another. The resulting

shape functions are piecewise-linear step functions defined on histogram bins. The hyperparameters subjected to Bayesian optimization include the number of bins (`max_bins`), the bin resolution for interaction terms (`max_interaction_bins`), the number of retained interaction pairs (`interactions`), the boosting step size (`learning_rate`), the minimum leaf size (`min_samples_leaf`), and number of boosting rounds (`max_rounds`). Because the EBM does not handle missing values natively, a scikit-learn pipeline with median imputation for numerical features and most-frequent imputation for categorical features was applied, fitted within training folds to prevent leakage.

IGANN. The *IGANN* model (Kraus et al., 2024), implemented via the `igann` library, replaces the piecewise-linear shape functions of the EBM with smooth curves learned by small Extreme Learning Machine (ELM) networks activated by an ELU nonlinearity. Each shape function f_j is represented by a two-layer network with a fixed random input projection and a regularized least-squares output layer, whose smoothness is controlled by the regularization parameter `elm_alpha`. The boosting process iteratively adds ELM units to minimize the residual, with the step size governed by `boost_rate`. This architecture allows IGANN to capture smooth, continuously differentiable non-linearities that the histogram-based EBM may only approximate in a staircase fashion, while retaining the fully additive and globally interpretable structure of Equation (C.1). Because ELMs are sensitive to input scale, a `StandardScaler` is applied upstream in the preprocessing pipeline. During hyperparameter search it was found that values of `elm_alpha` below 10^{-3} produced singular ELM matrices; the lower bound of the search space was therefore set to 10^{-3} to ensure numerical stability. The hyperparameters subjected to Bayesian optimization were the hidden layer width (`n_hid`), the number of boosting iterations (`n_estimators`), `boost_rate`, `elm_scale`, and `elm_alpha`.

D Feature Importance: XGBoost vs. LightGBM

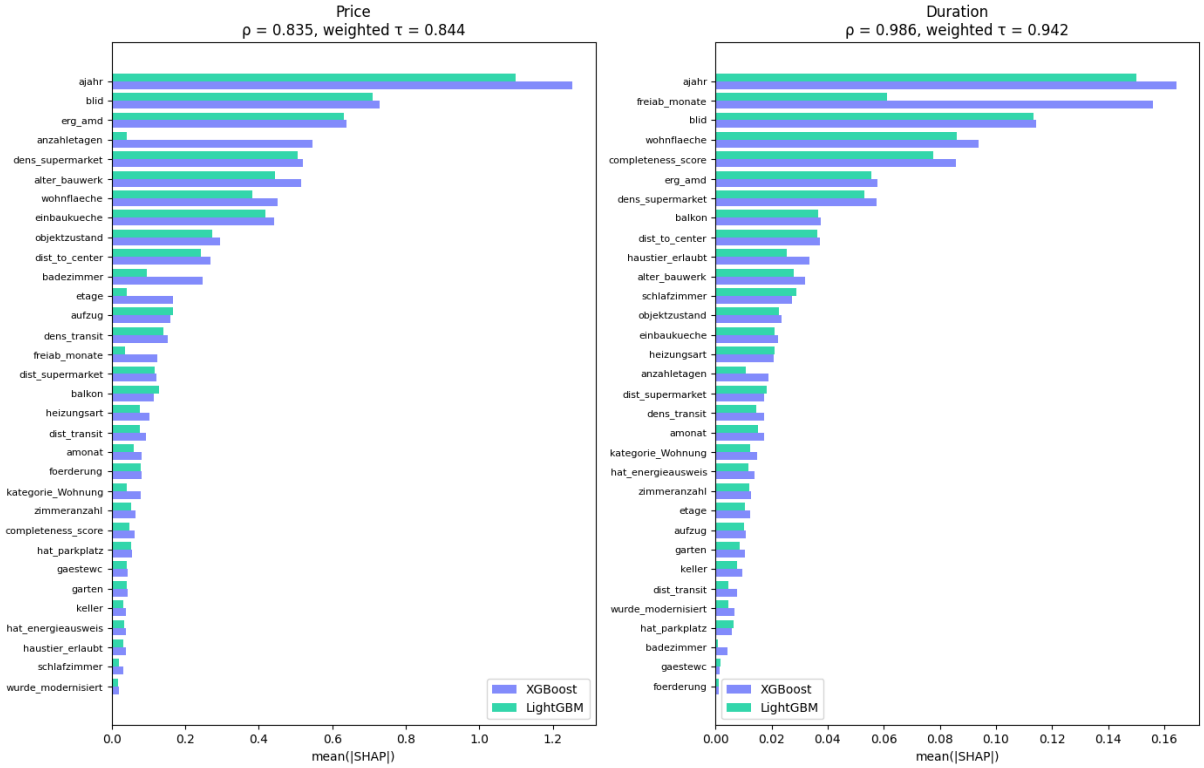


Figure D.1: Feature importance (mean absolute SHAP value) for XGBoost vs. LightGBM, separately for Price (Model A) and Duration (Model B). Spearman ρ and weighted Kendall's τ per target are given in the panel titles.

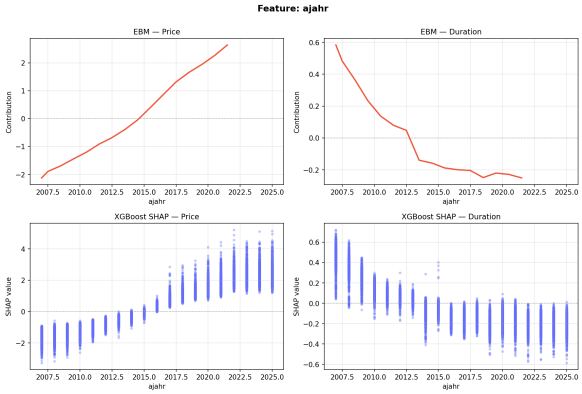
E Shape Functions: Classification and Plots

Table E.1: Cross-target classification of all features.

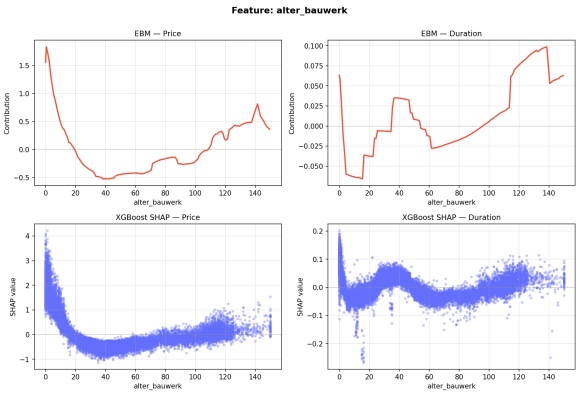
feature	cross_method	cross_target	notes
ajahr	agree	opposing	Reflects tightening rental market 2007–2025; rising prices coincide with shorter listing durations.
alter_bauwerk	agree	concordant	U-shape both targets; Neubau and Altbau segments show both price premium and longer listing duration, consistent with niche market segments.
amonat	partial	inconclusive	weak seasonal price signal; duration too noisy
anzahletagen	partial	concordant (weak)	weak positive effect both targets; EBM unreliable beyond 15 floors (sparse bins)
aufzug	agree	concordant	Elevator increases price and marginally increases listing duration; duration effect near negligible.
badezimmer	partial	inconclusive	clear, agreeing price effect; duration effect differs (EBM shows weak non-monotonic structure, XGBoost shows none)
balkon	agree	opposing	balcony increases price and decreases listing duration
blid	partial	opposing	nominal feature, curve interpretation limited; clear effects for single states
completeness_score	agree	opposing	counterintuitive: higher listing quality associates with lower price, longer duration

feature	**cross_method**	**cross_target**	**notes**
dens_supermarket	partial	opposing	agreement in core range (0–25); unstable in sparse tail (>25)
dens_transit	partial	opposing	Higher transit-stop density associates with lower price and longer duration — counterintuitive; possibly confounded with location type (dense transit corridors ≠ desirable residential microlocation)
dist_supermarket	agree	opposing	Clean, intuitive proximity effect: closer supermarket access commands price premium and shorter time-on-market.
dist_to_center	agree	opposing	Both center and periphery command price premiums with shorter duration; suburban mid-range is least desirable.
dist_transit	partial	inconclusive	erratic on both methods, weak signal overall
einbaukueche	agree	opposing	price premium and shorter duration
erg_amd	disagree	inconclusive	unclear feature semantics; extreme values misaligned between methods
etage	partial	inconclusive	Price effect agrees; Duration direction disputed between methods (EBM shows decreasing, XGBoost shows increasing trend).
foerderung	agree	opposing	Subsidized units: lower price, longer time-on-market.
freiab_monate	agree	concordant	Strong effect both targets; unusually large duration magnitude vs. other features.
gaestewc	partial	concordant	Both effects positive but negligible in magnitude; likely low practical relevance.
garten	agree	opposing	weak effect
hat_energieausweis	agree	opposing	Counterintuitive direction: having an energy certificate associates with lower price and longer duration; magnitude small.

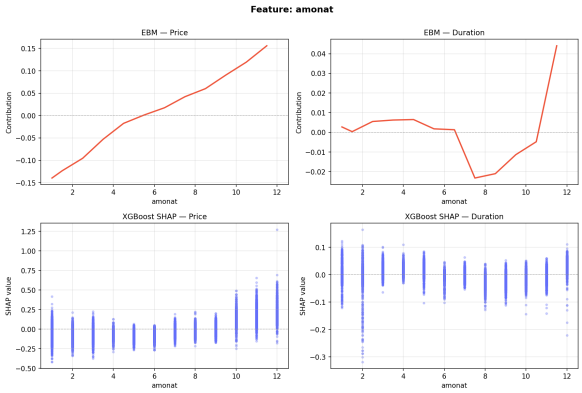
feature	cross_method	cross_target	notes
hat_parkplatz	agree	concordant	Both effects positive but duration magnitude negligible; price effect modest.
haustier_erlaubt	partial	mixed	Concordant for categories 0-1 (both lower for allowed pets); category 2 (negotiable) disputed: EBM shows opposing sign, XGBoost shows concordant
heizungsart	disagree	inconclusive	Nominal categorical feature; shape interpretation not meaningful.
kategorie_Wohnung	disagree	inconclusive	Nominal categorical feature; shape interpretation not meaningful.
keller	agree	opposing	Counterintuitive: cellar associates with lower price and longer duration.
objektzustand	partial	mixed	Price/Duration same sign in core range; EBM duration spike at high categories, not confirmed by XGBoost; unreliable at range extremes.
schlafzimmer	agree	concordant	Core range agrees both targets and models; tails sparse/unreliable
wohnflaeche	agree	mixed	small apartments and large units both command price premium, while the mainstream mid-size segment sees falling prices with rising time-on-market.
wurde_modernisiert	agree	opposing	Attractive amenity pattern like einbaukueche: price premium, shorter duration.
zimmeranzahl	agree	concordant	both curves move together across most of the range; brief sign divergences



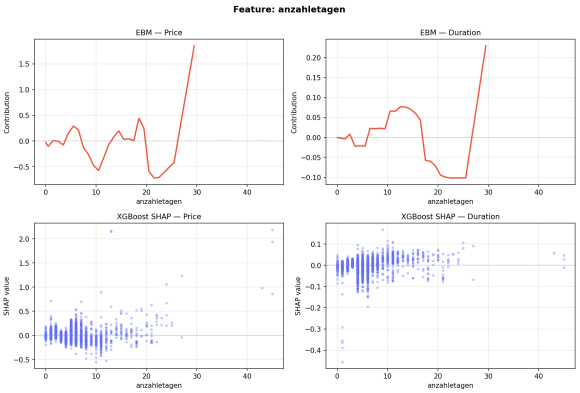
(a) ajahr



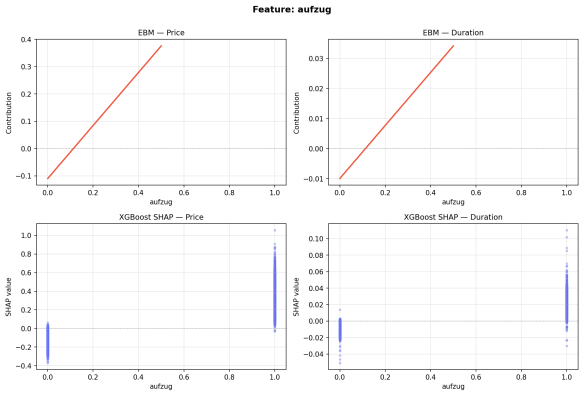
(b) alter_bauwerk



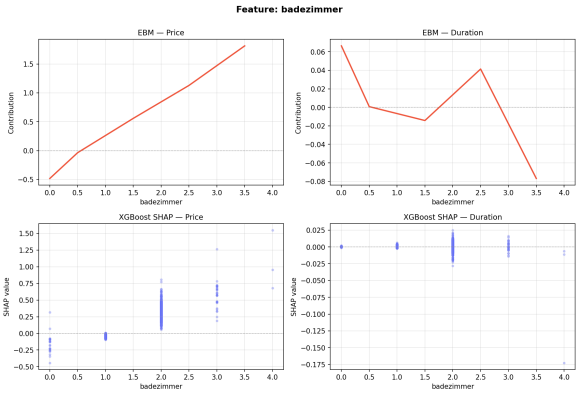
(c) amonat



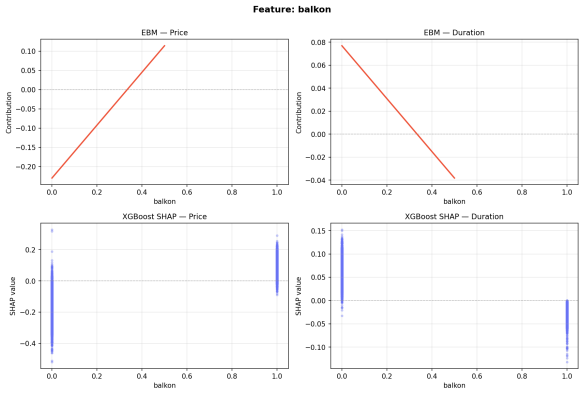
(d) anzahletagen



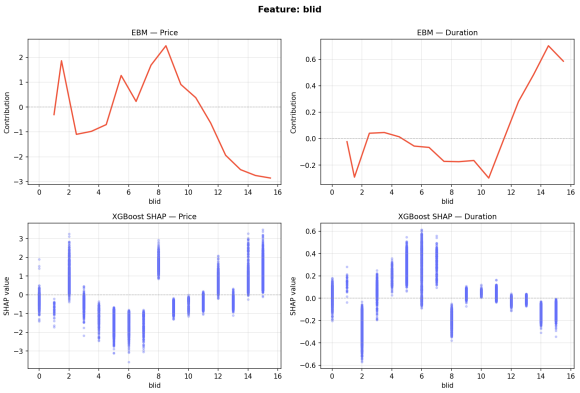
(e) aufzug



(f) badezimmer

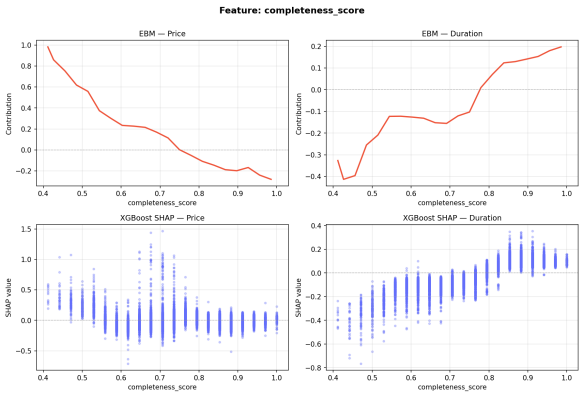


(g) balkon

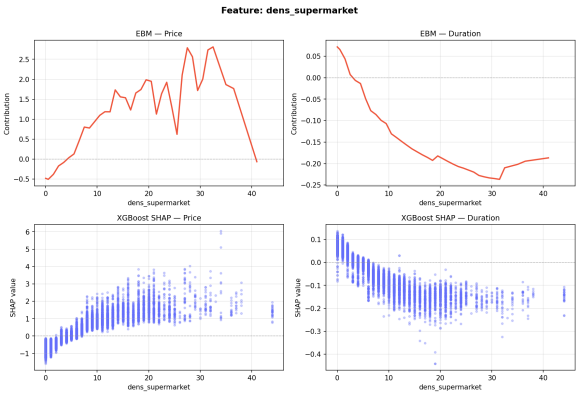


(h) blid

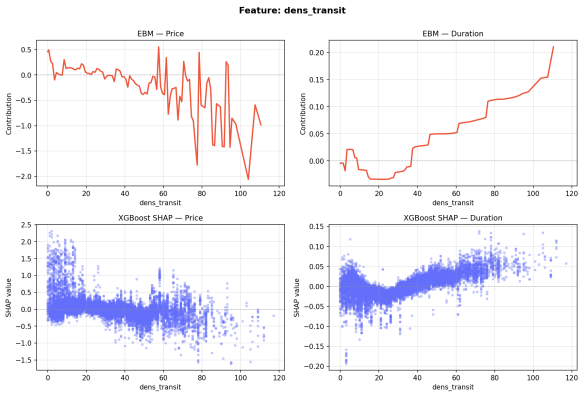
Figure E.1: Feature Plots (Teil 1)



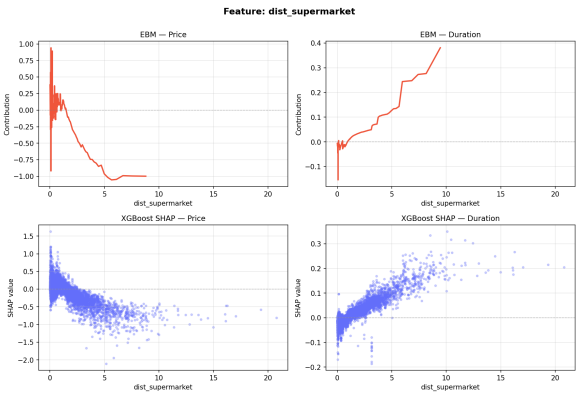
(a) completeness_score



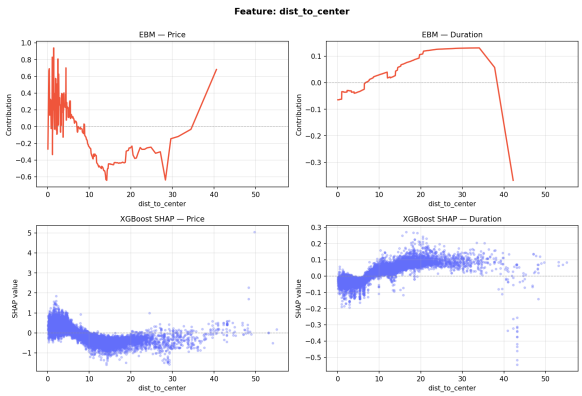
(b) dens_supermarket



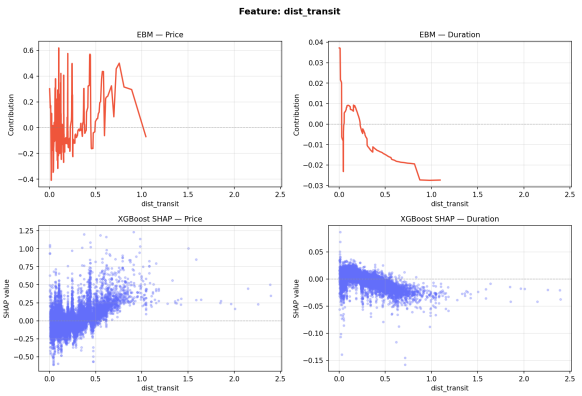
(c) dens_transit



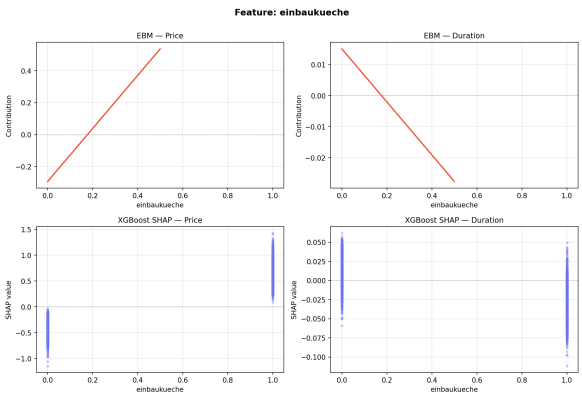
(d) dist_supermarket



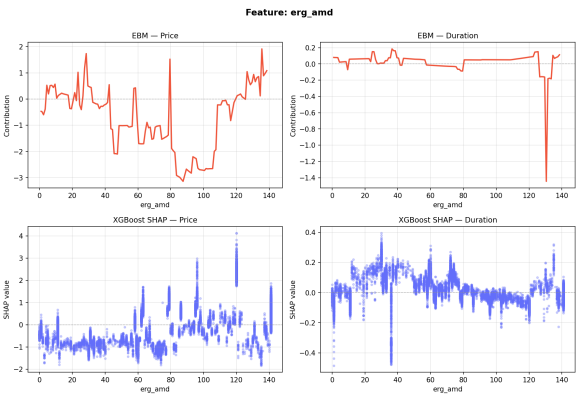
(e) dist_to_center



(f) dist_transit

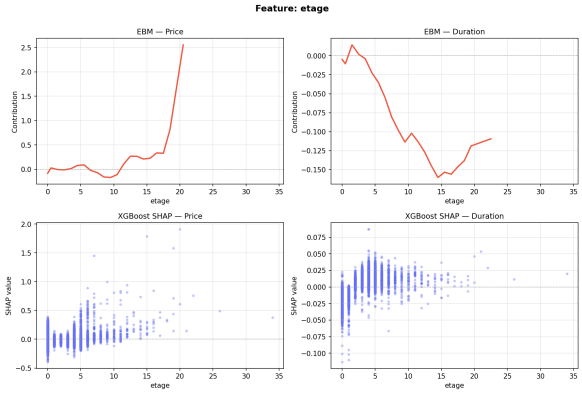


(g) einbaukueche

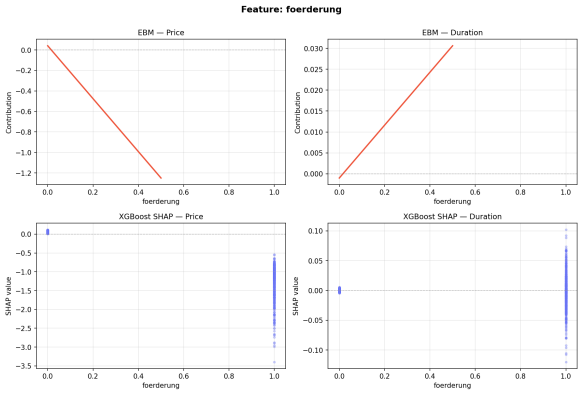


(h) erg_amd

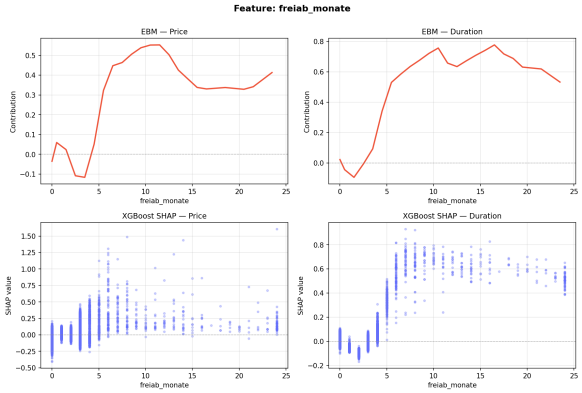
Figure E.2: Feature Plots (Teil 2)



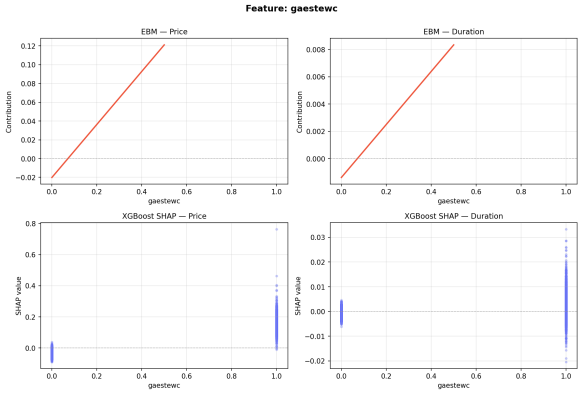
(a) etage



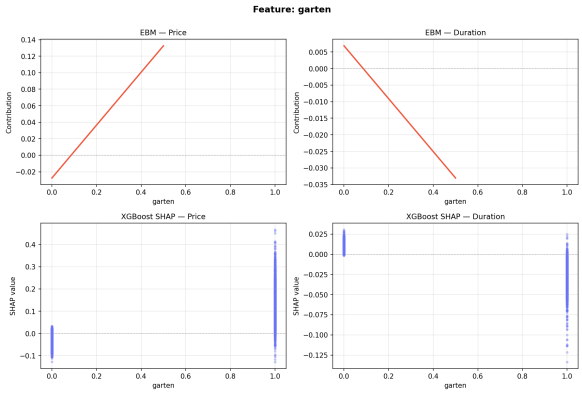
(b) foerderung



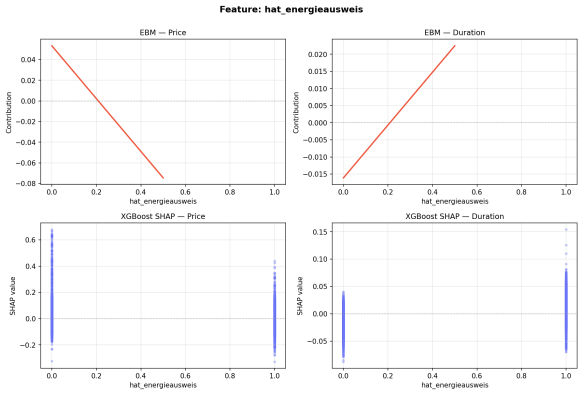
(c) freiab_monate



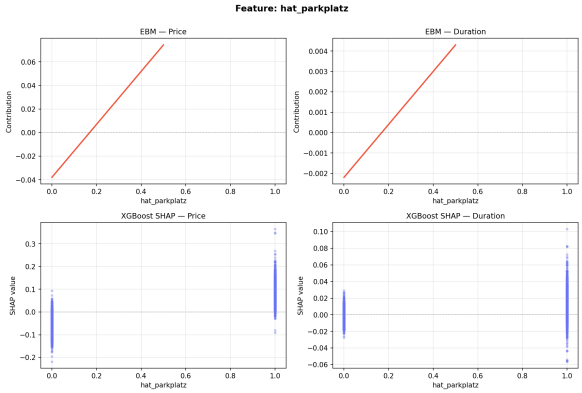
(d) gaestewc



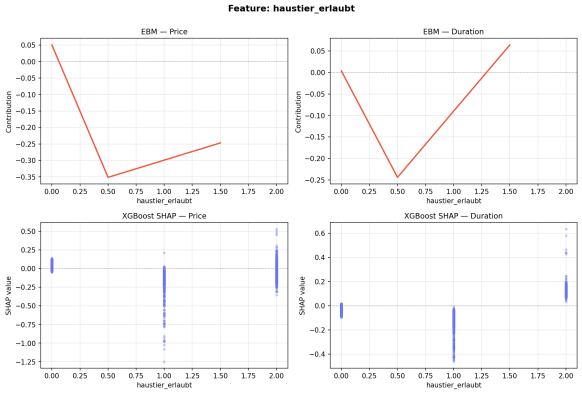
(e) garten



(f) hat_energieausweis

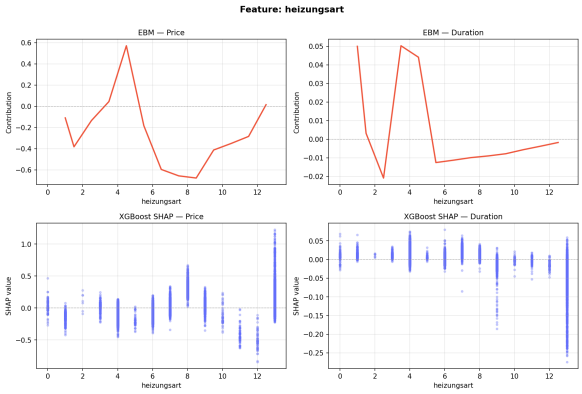


(g) hat_parkplatz

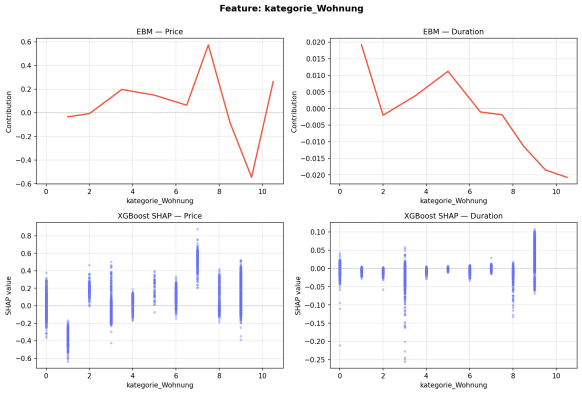


(h) haustier_erlaubt

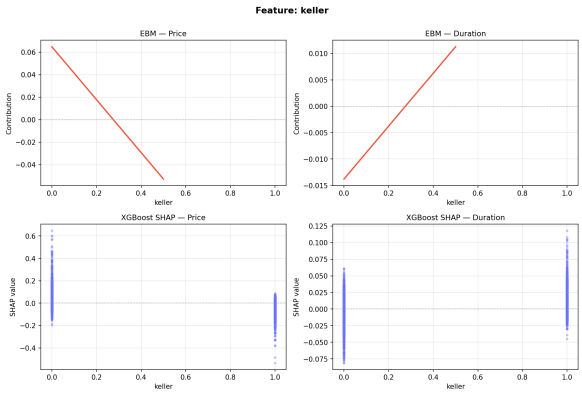
Figure E.3: Feature Plots (Teil 3)



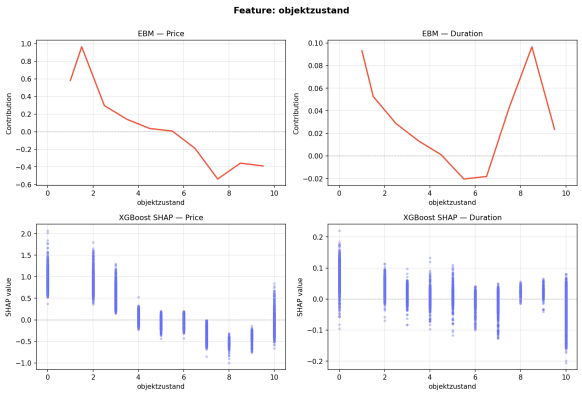
(a) heizungsart



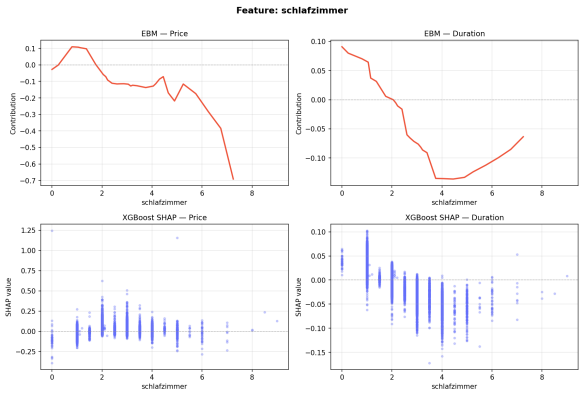
(b) kategorie_Wohnung



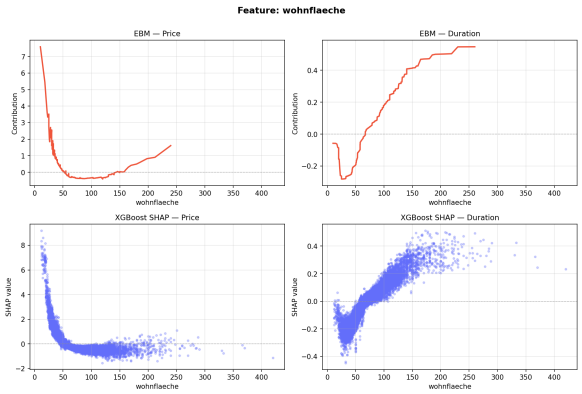
(c) keller



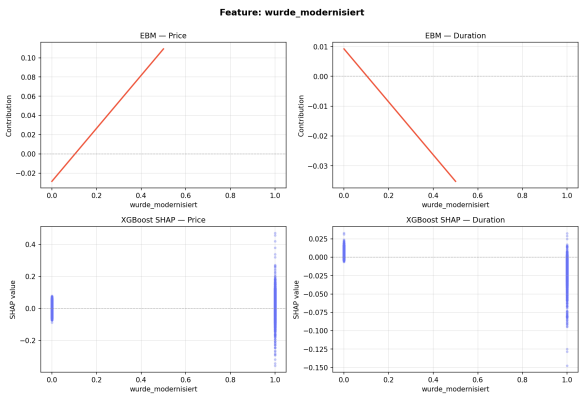
(d) objektzustand



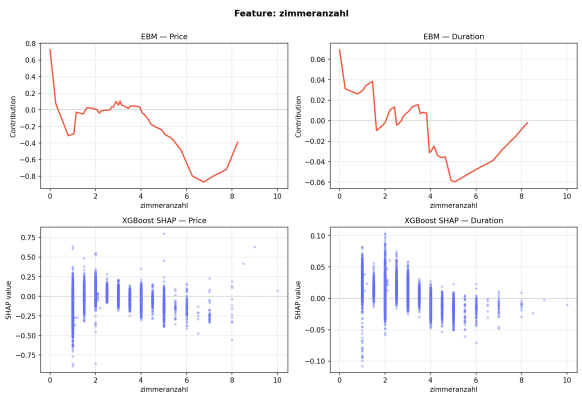
(e) schlafzimmer



(f) wohnflaeche



(g) wurde_modernisiert



(h) zimmeranzahl

Figure E.4: Feature Plots (Teil 4)

F Streamlit Application: Technical Documentation

Application Type: Multi-page Streamlit dashboard for model comparison, shape-function exploration, and apartment-level inference.

Component	Technical Specification
Entry point	`app/app.py` (wide layout, sidebar navigation)
Pages	`app/pages/1_Model_Comparison.py`, `2_Shape_Explorer.py`, `3_Apartment_Simulator.py`
Shared services	`app/utils.py` (model registry, metrics loading, shape-data loading, availability checks)
Model artifacts	`models/*.joblib`, metrics `models/metrics_model_A.csv`, `models/metrics_model_B.csv`
Shape artifacts	`data/processed/shapes_{igann,ebm}_{price,duration}.parquet`
Input schema source	`src/data/preprocessing.py` (`FEATURE_COLS`, numerical/categorical feature groups)
Inference path	Sidebar inputs → feature vector → optional imputers → model prediction
Explainability outputs	Global: precomputed shape functions (IGANN/EBM), Local: SHAP waterfall for XGBoost/LightGBM
Optional integration	ImmoScout24 listing scrape via `immo-scraper/scraper.py` (fallback: manual input)
Runtime	Python 3.11, Streamlit, Plotly, SHAP, Joblib, Pandas
Launch	`./start.sh` or `streamlit run app/app.py -server.port 8501`

Table F.1: Compact technical overview of the Streamlit application implementation.

UI Modules (compact).

- **Model Comparison:** renders MAE/RMSE/R^2 tables and MAE ranking bar charts per target.
- **Shape Explorer:** feature-wise IGANN/EBM effect curves for `price` and `duration`.
- **Apartment Simulator:** user-defined apartment profile, dual-target prediction, model-specific explanation output.

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